

EA Prerequisite: Fundamental Commissioning and Verification

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Commissioning Process Scope

Complete the following commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies, in accordance with ASHRAE Guideline 0-2013 and ASHRAE Guideline 1.1-2007 for HVAC&R Systems, as they relate to energy, water, indoor environmental quality, and durability.

- ▶ Develop the OPR.
- ▶ Develop a BOD.

The commissioning authority (CxA) must do the following:

- ▶ Review the OPR, BOD, and project design.
- ▶ Develop and implement a Cx plan.
- ▶ Confirm incorporation of Cx requirements into the construction documents.
- ▶ Develop construction checklists.
- ▶ Develop a system test procedure.
- ▶ Verify system test execution.
- ▶ Maintain an issues and benefits log throughout the Cx process.
- ▶ Prepare a final Cx process report.
- ▶ Document all findings and recommendations and report directly to the owner throughout the process.

Requirements for exterior enclosures are limited to inclusion in the owner's project requirements (OPR) and basis of design (BOD), as well as the review of the OPR, BOD and project design. ASTM E2947-16: Standard Guide for Building Enclosure Commissioning provides additional guidance.

The review of the exterior enclosure design may be performed by a qualified independent member of the design or construction team (or an employee of that firm) who is not directly responsible for design of the building enclosure for the project.

Commissioning Authority Qualifications

By the end of the design development phase, engage a commissioning authority with the following qualifications.

- ▶ The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- ▶ The CxA may be a qualified employee of the owner, an independent consultant, an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.
 - For projects smaller than 20,000 square feet (1 860 square meters), the CxA may be a qualified member of the design or construction team.

In all cases, the CxA must report his or her findings directly to the owner.

Current Facilities Requirements and Operations and Maintenance Plan

Prepare and maintain a current facilities requirements and operations and maintenance plan that contains the information necessary to operate the building efficiently. The plan must include the following:

- ▶ a sequence of operations for the building;
- ▶ the building occupancy schedule;
- ▶ equipment run-time schedules;
- ▶ setpoints for all HVAC equipment;
- ▶ set lighting levels throughout the building;
- ▶ minimum outside air requirements;
- ▶ any changes in schedules or setpoints for different seasons, days of the week, and times of day;
- ▶ a systems narrative describing the mechanical and electrical systems and equipment;
- ▶ a preventive maintenance plan for building equipment described in the systems narrative; and
- ▶ a commissioning program that includes periodic commissioning requirements, ongoing commissioning tasks, and continuous tasks for critical facilities.

Data Centers only

For small projects with computer room peak cooling loads less than 2,000,000 Btu/h (600 kW) or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA may be a qualified employee of the design or construction team.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Updated referenced standards.

Further Explanation

Commissioning Authority Qualifications

Delete the phrase "If the project is pursuing fundamental commissioning only" in the second paragraph.

In Table 1. Who can be the CxA:

- ▶ Revise the heading "be CxA for..." to "be CxA" in the right column
- ▶ Delete the subheading "fundamental Cx" in the right column
- ▶ Delete subheading "enhanced Cx" and all rows beneath it in the far right column of the table

Related Credit Tips

Refer to the LEED v4 reference guide, with the following modification:

- ▶ EA credit Renewable Energy. Renewable energy systems installed on-site must be commissioned under this prerequisite.

Referenced Standards:

- ▶ ASHRAE Guideline 0-2013, The Commissioning Process
- ▶ ASHRAE Guideline 1.1-2007, HVAC&R Technical Requirements for the Commissioning Process
- ▶ ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Testing building systems after installation is fundamental to ensuring that systems function as designed. The development of a commissioning plan ensures that the building owner and facility managers have the information necessary to operate the building efficiently.

EA Prerequisite: Minimum Energy Performance

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To promote resilience and reduce the environmental and economic harms of excessive energy use and greenhouse gas emissions that disproportionately impact frontline communities by achieving a minimum level of energy efficiency for the building and its systems.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Comply with ANSI/ASHRAE/IESNA Standard 90.1-2016, with errata or a USGBC-approved equivalent standard.

ASHRAE 90.1-2016 Compliance pathways in Section 4.2.1.1 include compliance with all mandatory provisions, and compliance with one of the following:

- ▶ Prescriptive provisions of Sections 5 through 10
- ▶ Section 11 *Energy Cost Budget Method*
- ▶ Normative Appendix G *Performance Rating Method*. When using Appendix G, the Performance Cost Index (PCI) shall be less than or equal to the Performance Cost Index Target (PCI_t) in accordance with the methodology provided in Section 4.2.1.1. Document the PCI, PCI_t, and percentage improvement using metrics of cost or greenhouse gas (GHG) emissions.
- ▶ Exception to Mandatory Measures requirements: For ASHRAE 90.1-2016 mandatory provisions where the Appendix G *Performance Rating Method* provides a methodology for demonstrating savings between the Proposed Building Performance (PBP) and the Baseline Building Performance (BBP), projects may model the Proposed Building Performance as designed in lieu of compliance with the mandatory provisions.
- ▶ Exceptional Calculations modeled in accordance with Section G2.5 may be modeled to document minimum prerequisite compliance.
- ▶ Only on-site or on-campus renewable energy that meets ASHRAE Standard 90.1-2016 Section G 2.4.1 requirements for on-site renewable energy may be used to meet ASHRAE Standard 90.1-2016 performance requirements.

GUIDANCE

The following guidance addresses both EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance since the requirements and LEED documentation are closely linked for the Prerequisite and Credit.

Refer to the LEED v4 reference guide, EA Prerequisite Minimum Energy Performance for the referenced sections.

Beta Update

Updated referenced standards and added a new greenhouse gas emissions metric ensure that LEED continues to be a global leadership standard for energy performance and encourage owners to directly consider and address building carbon emissions.

Step-by-Step Guidance

ASHRAE Standard 209 provides a step-by-step methodology for applying energy modeling to inform the design process. Project teams are encouraged, though not required, to apply the guidance in ASHRAE Standard 209 as a best-practice approach for informing design through energy modeling. Following the guidance in Standard 209 will help project teams document achievement of LEED EA prerequisite Minimum Energy Performance Prerequisite, EA credit Optimize Energy Performance, and the energy modeling requirements for IP credit Integrative Process.

Step 1. Determine climate zone

Identify the project's climate zone according to ASHRAE 90.1-2016, Annex 1 (see *Further Explanation, Climate Zone Determination*).

Step 2. Review and address ASHRAE mandatory requirements

Early in the design process, review the mandatory provisions of ANSI/ASHRAE/IESNA Standard 90.1-2016, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.). Read through Sections 5.4, 6.4, 7.4, 8.4, 9.4, and 10.4 to understand how the building design must respond to these requirements. Many mandatory requirements can easily be incorporated to the project in early design but are much harder to incorporate later in design and/or during construction.

- ▶ Typically, the architect is responsible for Section 5.4, Building Envelope; the mechanical engineer and plumbing designer are responsible for Sections 6.4, HVAC, and 7.4, Service Water Heating; and the electrical engineer is responsible for Sections 8.4, Power, and 9.4, Lighting. Compliance with Section 10.4 requires coordination across multiple disciplines.
- ▶ Ensure that the project complies with the mandatory measures throughout the design, construction, and commissioning process, particularly when major design decisions are implemented.
- ▶ Confirm that compliant components are included in the final construction documents.
- ▶ If compliance with ASHRAE 90.1-2016 mandatory provisions will be a hardship for the project, and the project intends to demonstrate compliance using the Appendix G Performance Rating Method, identify whether Appendix G Performance provides a method for quantifying savings of these mandatory provisions in the Proposed design as compared to the Baseline. For mandatory measures where Appendix G provides a methodology for demonstrating savings between the Proposed Building Performance (PBP) and the Baseline Building Performance (BBP), in lieu of compliance with the mandatory provisions, the Proposed Building efficiencies and controls may be modeled as designed, while the Baseline Building efficiencies and controls are modeled consistent with ASHRAE 90.1-2016 Appendix G requirements. Examples include:
 - In lieu of complying with mandatory daylighting and occupancy sensor control requirements, document the proposed model without credit for lighting controls, with lighting schedules modeled identically to the baseline model.
 - If the exterior lighting power allowance exceeds the mandatory allowance, model the proposed exterior lighting power as designed, and model the Baseline consistent with Appendix G.
 - If the HVAC efficiency does not meet the mandatory provisions, model the proposed efficiency as designed, and model the Baseline consistent with Appendix G.
- ▶ The following additional exceptions to mandatory measures may also be applied:
 - Section 5.4.3.4 Vestibules: The project may apply ASHRAE 90.1-2016 Addendum bf, Section 5.4.3.4, which adds exceptions for:

- Self-closing doors in buildings in Climate Zone 0, 3, and 4 that have an air curtain complying with Addendum bf, Section 10.4.5.
- Self-closing doors in buildings 15 stories or less in Climate Zones 5 through 8 that have an air curtain complying with Addendum bf, Section 10.4.5.

Note that the air curtain energy must be included in the Proposed design model and not in the Baseline if applying the Appendix G Performance Rating Method.

- Section 8.4.2 Automatic Receptacle Control:
 - Path 1. Projects Using the Appendix G Performance Rating Method may model a penalty in the energy model for the spaces where mandatory ASHRAE 90.1 receptacle controls are not implemented. The following modeling requirements apply:
 - The Proposed receptacle power density modeled for these spaces shall be the greater of 0.75 Watts per square foot (8.1 Watts per square meter) or the design coincident peak receptacle power density (if known)
 - The receptacle schedule modeled in the Proposed design for these spaces shall have a minimum Equivalent Full Load Hours of operation no less than: the ASHRAE 90.1-2016 User's Manual default schedule for office occupancy (2,920 Equivalent Full Load Hours per year); or 120% of the occupied hours of operation for the facility; or detailed justification shall be provided supporting an alternate schedule.
 - The Proposed model shall include either a 20% increase in the receptacle power density for these spaces OR a 20% increase in the scheduled receptacle Equivalent Full Load Hours of Operation versus the Baseline model.
 - Path 2: Projects must demonstrate that the project has implemented efficiency measures that will achieve an equal or greater reduction in receptacle energy consumption and will persist for a similar timeframe to those achieved by ASHRAE 90.1-2016 Section 8.4.2. It is recommended that a Credit Interpretation Request be submitted when pursuing this approach. The project must provide documentation regarding the receptacle equipment controls that will be implemented for the project; and must provide justification supporting the claim that the savings over the life of the efficiency measure will be similar to those anticipated for a project compliant with Section 90.1-2016 Section 8.4.2. Note: the project is not eligible for any further receptacle savings in these spaces using the Appendix G Exceptional Calculation Method or using the Prescriptive path for measures relying on ENERGY STAR eligible equipment features.

(See Further Explanation, Table 1. Changes in ASHRAE 90.1 mandatory requirements, 2010 to 2016).

Step 3. Identify energy use target for building

This step is required for all projects pursuing credit under IP credit Integrative Process and recommended for all other projects.

Set an energy goal for the project early in the design process. Identifying an energy goal can help prioritize efficiency strategies, integrate systems, reduce first costs, and improve building performance.

For IP Credit Integrative Process, the target must be established using one of the following metrics:

- ▶ kBtu per square foot-year (kWh per square meter-year) of site energy use
- ▶ kBtu per square foot-year (kWh per square meter-year) of source energy use
- ▶ pounds per square foot-year (Kg per square meter-year) of greenhouse gas emissions
- ▶ energy cost per square foot-year (cost per square meter-year)

For building types such as manufacturing, if a different metric is more appropriate for benchmarking building energy consumption (e.g. kBtu per pound of finished product (kWh per kilogram of finished project)), project teams may use that metric in lieu of the metrics above. When using a different metric, provide a brief narrative supporting that the metric used is a more appropriate means of benchmarking building energy consumption for the building type and function.

Consider using ENERGY STAR's Target Finder to develop the EUI goal that will meet the credit requirements.

Consider applying the guidance in ASHRAE Standard 209 Section 5.4 and Informative Appendix B (Benchmark Information) when establishing the energy goal for the project.

Step 4. Select option for credit compliance.

Select the appropriate option in EA credit Optimize Energy Performance for the project (see *Further Explanation, Selecting an Option*). Review the requirements for EA credit Optimize Energy Performance before making a selection.

- ▶ Option 1. Energy Performance Compliance is available to all projects. This option is the best method for informing design decisions throughout the design process, and has the greatest number of points available under EA credit Optimize Energy Performance. For projects using this method, a Baseline Building Performance Model and Proposed Building Performance model are developed consistent with ASHRAE 90.1-2016 Appendix G, Performance Rating Method.
- ▶ Options 2 and 3 are for projects intending to apply simple upgrades to mechanical, envelope, lighting, appliances, and/or process equipment. Projects with minimal scope, such as Core and Shell projects with scope limited to the building envelope and exterior lighting often apply this option, since prerequisite compliance may be documented based solely on the project scope, and Optimize Energy Performance credit may be documented for the specific elements that are within the project scope.

Projects must demonstrate compliance with EA prerequisite Minimum Energy Performance using the ASHRAE 90.1-2016 prescriptive compliance pathway to apply these options. Projects pursuing this option should work with the architect and engineers to assess the prescriptive requirements of ANSI/ASHRAE/IESNA Standard 90.1-2016, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.) and ensure that the design will comply with envelope, HVAC, service water-heating, and lighting requirements, per Sections 5.5, 6.5, 7.5, 9.2.2 for all elements within the project scope of work. Compliance with prescriptive ASHRAE 90.1-2016 requirements and prescriptive EA credit Optimize Energy Performance requirements should be verified early in the design process, with ongoing verification of compliance throughout the design and construction process.

- Option 2: Project teams may pursue a limited number of points under EA credit Optimize Energy Performance. The eligible project types for Option 2 include the following:
 - Small to medium office buildings, less than 100,000 square feet (9 290 meters)
 - Medium to large box retail buildings, 20,000 to 100,000 square feet (1 860 to 9 290 square meters)
 - School buildings, any size
 - Large hospitals, more than 100,000 square feet (9 290 square meters)
 - Grocery stores
- Option 3 is a prescriptive option available for projects with less than 2,000 square feet (186 square meters) of data center space, laboratory space, or manufacturing space.

- ▶ For projects using the BD+C Data Center rating system, Option 4 may be used to demonstrate system optimization of the Data Center Mechanical and Electrical equipment using ASHRAE 90.4 2016.
- ▶ If the project is not pursuing any points under EA credit Optimize Energy Performance, the project may demonstrate EA prerequisite Minimum Energy Performance compliance using ASHRAE 90.1-2016 Section 11 Energy Cost Budget. This option uses energy modeling with trade-offs but has different Baseline building modeling requirements than the normative Appendix G performance rating method.
- ▶ Refer to the LEED Credit Library *Pilot alternative compliance paths (ACP)* for additional options for demonstrating EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Pilot ACPs test credits that achieve a similar intent to the referenced credit by applying a new strategy. For EAp Minimum Energy Performance and EAc Optimize Energy Performance, these may include alternative metrics for evaluating decarbonization and energy efficiency using ASHRAE 90.1-2016 Appendix G, entirely new compliance paths intended to streamline the analysis and documentation process, or compliance paths that reference alternative standards.

Step 5. Develop preliminary energy model or alternate energy analysis

To achieve EA credit Optimize Energy Performance, project teams must analyze efficiency measures during the design process, focusing on load reduction and HVAC-related strategies or passive measures appropriate for the facility, and account for the results during design decision making.

For projects using Option 1. Energy Performance Compliance, the best approach for analyzing efficiency measures is a preliminary energy model, which evaluates heating and cooling load reduction strategies, passive HVAC strategies, and HVAC efficiency and control strategies (see *Further Explanation, Developing a Preliminary Energy Model* and *Further Explanation, Modeling HVAC Systems*). ASHRAE Standard 209 provides a standardized methodology which may be used for developing preliminary energy models that are used to inform the design process (See ASHRAE 209 Sections 6.3 - Load Reduction Modeling and 6.4 - HVAC System Selection Modeling).

- ▶ Developing an early model of the proposed design will help the design team explore the energy consequences of design options and will provide an early estimate of energy performance.
- ▶ When evaluating energy usage in different scenarios, consider strategies for lighting and daylighting, envelope, orientation, and passive conditioning and ventilating systems, in terms of projected energy savings and capital costs as they relate to all building systems. If pursuing the Integrative Process Credit, evaluate these parameters at a concept level early in design.

Project teams may also use past energy analyses of similar buildings or published energy modeling results, such as the ASHRAE Advanced Energy Design Guides (AEDGs) to guide decision making in lieu of a preliminary energy model, though the results will be less project-specific. The AEDGs were designed around specific building types and sizes by climate zone, making the recommendations most appropriate for projects with attributes similar to those specified types, sizes, and locations.

Step 6. Ongoing iterations of Design Phase Energy Model (Option 1. Energy Performance Compliance), or Prescriptive compliance documentation (Options 2 or 3).

Option 1. Energy Performance Compliance

For projects pursuing EA credit Optimize Energy Performance Option 1. Energy Performance Compliance:

Once the HVAC system and other design parameters are established, build or update the proposed building energy model to reflect the anticipated design (see *Further Explanation, Building the Proposed Energy Model*).

- ▶ Update the proposed model to reflect changes that occur throughout the design process to optimize energy performance and assist with design decisions.
- ▶ Ensure that all efficiency strategies are analyzed well before design documents are finalized.
- ▶ For elements or systems that cannot be readily modeled by the software or to document credit for unregulated loads, use the Exceptional Calculation Method (see *Further Explanation, Exceptional Calculation Method* and v4 Reference Guide, *Further Explanation, Common Issues with Energy Modeling*).
- ▶ Document credit for qualifying renewable energy in the proposed energy model (see *Further Explanation, Applying Renewable Energy Savings*).

For projects pursuing EA credit Optimize Energy Performance, Option 1. Energy Performance Compliance:

Build a baseline model that reflects the minimum requirements according to ASHRAE 90.1-2016, Appendix G (see *Further Explanation, Building the Baseline Performance Model*).

- ▶ When modifications are made to the proposed energy model, update the baseline accordingly.
- ▶ Consider constructing the baseline model early in the design process so that the design team can see the effect of design changes on the percentage savings relative to ASHRAE 90.1. This will contribute toward achieving more points under the related credit.
- ▶ Use the Minimum Energy Performance Calculator to help create the baseline model. This tool was designed to help project teams create a baseline model in alignment with Appendix G requirements.

Update the proposed energy model as necessary to reflect final construction details and specifications and make any necessary corresponding updates to the baseline model.

Use the results from the baseline and proposed models and the Building Performance Factor to determine the anticipated energy cost and greenhouse gas emissions savings (see *Further Explanation, Calculations, Energy Cost* and *Greenhouse Gas Emissions*). Either the cost or the GHG emissions metric may be used to show prerequisite compliance.

Prerequisite Compliance Only – ASHRAE 90.1-2016 Section 11 Energy Cost Budget

For projects that are not pursuing EA credit Optimize Energy Performance, and are documenting compliance using ASHRAE 90.1-2016 Section 11, Energy Cost Budget, complete the ASHRAE 90.1 Section 11 design energy cost and energy cost budget models, and complete the ASHRAE 90.1 ECB forms demonstrating compliance.

Option 2, 3, or 4. Prescriptive Compliance

Prepare final ASHRAE 90.1 documentation confirming compliance with the mandatory and prescriptive requirements of ASHRAE 90.1-2016.

For projects pursuing Optimize Energy Performance, see Step-by-Step Guidance, Prescriptive Compliance.

Further Explanation

Calculations

For projects using ASHRAE 90.1-2016 Appendix G, Performance Rating Method, the following equations apply:

Equation 1. Section G1.2.2: Performance Cost Index

Performance Cost Index = Proposed building performance / Baseline building performance.

where Proposed building performance and Baseline building performance are calculated in accordance with ASHRAE 90.1-2016 Appendix G.

Equation 2. Section 4.2.1. Performance Cost Index

$$PCI_t = [BBUEC + (BPF \times BBREC)] / BBP$$

where:

PCI = Performance Cost Index calculated in accordance with ASHRAE 90.1-2016 Section G1.2 as described above.

BBUEC = Baseline Building Unregulated Energy Cost, the portion of the annual energy cost of a baseline building design that is due to unregulated energy use.

BBREC = Baseline Building Regulated Energy Cost, the portion of the annual energy cost of a baseline building design that is due to regulated energy use.

BPF = Building Performance Factor from Table 4.2.1.1. For building area types not listed in Table 4.2.1.1 use "All others." Where a building has multiple building area types, the required BPF shall be equal to the area-weighted average of the building area types. The Building Performance Factors from Table 4.2.1.1 (BPF) represent the average ratio of ASHRAE 90.1-2004 versus ASHRAE 90.1-2016 regulated energy cost for a given building type and climate. For example, a BPF of 0.59 represents an ASHRAE 90.1-2016 regulated energy cost that is 59% of the 90.1-2004 regulated cost for the given building type and climate. Table 4.2.1.1 lists

BBP = Baseline Building Performance.

Equations 1 and 2 adjustments for greenhouse gas emission metric:

When using Greenhouse gas emissions as the metric, replace all ASHRAE 90.1-2016 references to cost with Greenhouse Gas Emissions (CO₂ equivalent emissions). Rather than using utility rates, use the appropriate greenhouse gas emission coefficients for each energy source (See [*Further Explanation. Greenhouse Gas Emissions*](#))

Total LEED points for EAc Optimize Energy Performance are determined by summing the points documented in Table 1 (cost metric) PLUS the points documented in Table 2 (GHG emissions metric), where points for each Table are determined based on the project percent improvement PCI below the PCI_t documented for each metric.

For Table 1:

- Percent cost PCI below PCI_t = 1-PCI/PCI_t

For Table 2, calculate PCI and PCI_t using greenhouse gas emissions instead of cost.

- Percent GHG Emissions PCI below PCI_t = 1-PCI/PCI_t

If a project consists of a combination of New Construction and either Major Renovation or Healthcare, use Equation 1 to determine the appropriate percentage improvement target from Table 1 or Table 2 (points for percentage improvement in energy performance).

Equation 3. Target energy savings for combination of New Construction, Major Renovation, and/or Healthcare (HC).

Target percentage = {(existing or HC floor area / total floor area) x target percentage of savings for Major Renovation or Healthcare} + {(new non-HC floor area / total floor area) x target percentage of savings for New Construction}

Climate Zone Determination

Determining the right climate zone for the project is essential since the requirements are specific to each climate zone. ASHRAE 90.1-2016 defines eight climate zones (Miami is in climate zone 1; Anchorage is in climate zone 8) and three climate types: A (moist), B (dry), and C (marine).

To find the project's climate zone and type, consult ASHRAE 90.1-2016, Annex 1. For projects in the U.S, refer to the appropriate state and county in Table Annex 1-1. For projects in Canada, refer to the province and location in Table Annex 1-2. For locations outside of the U.S. and Canada, refer to the closest or most similar location in Table Annex 1-3. International projects may also refer to ASHRAE Standard 169-2020 to determine the project's climate zone based on historical weather data for the project's location.

Selecting an Option

Determining which option is most appropriate for the project requires knowing the extent of energy performance feedback desired during the design process.

- ▶ If detailed feedback is important during the design process, or the project is targeting a high level of energy performance and low greenhouse gas emissions, then the performance option (1) is most appropriate. Energy modeling generates information on the potential savings associated with various efficiency measures, both in isolation and in combination with other measures. Often this includes estimates of overall energy use, greenhouse gas emissions, or cost savings for the project, which can help gauge progress toward an energy savings and greenhouse gas emissions goal or achievement of points under the related credit.
- ▶ If the owner or design team requires only limited feedback, and the project is not targeting significant energy savings then one of the prescriptive options (2, 3, or for Data Centers - 4) may be more appropriate. These options are best suited for projects with standard systems and provide only limited feedback, in that all efficiency measures must be incorporated to achieve the prescribed threshold for energy performance.

Performance Path

The following factors could indicate that Option 1 would be advantageous to the project:

- ▶ The project is targeting a high level of energy performance
- ▶ None of the Optimize Energy Performance prescriptive pathways are available to the project because of the building's type or size.
- ▶ The project has an HVAC system that is not covered by one of the prescriptive options.
- ▶ The project team wants to explore the energy performance and load reduction effects of several envelope and lighting designs and mechanical systems.
- ▶ The project team is planning to maximize the number of points available through EA credit Optimize Energy Performance.
- ▶ The project team wants to achieve efficiency trade-offs between systems, offsetting the lower efficiency of one system by the improved efficiency of another.
- ▶ The owner is interested in commercial building federal tax credits or state, local, or utility incentives that require energy modeling. The modeling requirements for such incentive programs may be different from the ASHRAE 90.1-2016 requirements, however.
- ▶ The owner wants an estimate of the carbon reductions or lower operating costs (energy savings, demand charge savings) from energy strategies, beyond a simple calculation for individual energy conservation measures.

Project teams pursuing Option 1 should consider referencing ASHRAE Standard 209-2018, Energy Simulation Aided Design for Buildings except Low Rise Residential Buildings, which defines best

practices and minimum requirements for providing energy design assistance using building energy simulation and analysis.

Before undertaking energy modeling as part of the performance path, consider the timing of the simulation preparation and presentation, and understand the costs and benefits of energy modeling as it relates to the project. When energy modeling is conducted late in design, its value is very limited, except as a compliance tool: the model can only estimate the energy savings of the design.

In contrast, if initiated early and updated throughout the design process, energy modeling can be a decision-making tool, giving feedback as part of the larger analysis of building systems and components. The best value will be seen when energy modeling is used as a tool in an integrated design process because it enables a more informed, cost-effective selection of efficiency strategies.

Note: Early design phase analysis is required to earn points under EA credit Optimize Energy Performance.

Develop clear expectations for the presentations of modeling results and their integration into the project schedule. Ideally, iterations of the model will be presented to the team during each stage of design, beginning as early as possible, when the project goals are incorporated into preliminary plans. Updates should be presented as the design is developed further to incorporate engineering and architectural details, and again when the construction documents are being prepared.

Regardless of the project design phases, energy modeling can still be performed as the design progresses. However, the potential benefit of energy modeling decreases as the design becomes finalized and opportunities for incorporating changes are lost. Ask the project's energy modeler to provide a schedule that integrates energy modeling into the design process, with appropriate milestones.

To develop an accurate and compliant energy model, it is important that the energy modeler read and understand ASHRAE 90.1-2016 (Appendix G and all mandatory measures from each Section in particular) in its entirety, not just the portions that apply to the project. This will enable a more complete understanding of the energy modeling protocols and methodologies required for LEED projects (see *Further Explanation, ASHRAE 90.1, 2016 versus 2010*). The energy modeler should also consider reading the ASHRAE 90.1-2016 User's Manual, which provides examples and further guidance relevant to Appendix G.

Prescriptive paths

The following factors could indicate that Option 2 or Option 3 would be advantageous to the project:

- ▶ The project budget and timeline would benefit from simplified decision making and analysis during the project design.
- ▶ The additional cost of energy modeling would not be warranted.

Although the prescriptive paths are applicable to some large or complex projects, such as hospitals, they were designed primarily for smaller projects, for which the cost of energy modeling would represent a high percentage of the project budget.

The prescriptive paths are available only for projects that meet certain criteria. Review the project's eligibility for the ASHRAE 50% Advanced Energy Design Guides and/or Option 3. Systems Optimization, and Option 4: Data Centers Only – System Optimization. If these prescriptive option do not fit the project type, the team must pursue Option 1 in order to achieve points under EA credit Optimize Energy Performance.

If the project is eligible for both of the prescriptive options, determine which is more appropriate based on the specific option requirements as well as future credit goals. The building type, for example, may not match those in the AEDGs, or the Option 3 prescriptive requirements may align better with the project's goals and design.

Option 2. ASHRAE 50% AEDG, delivers a 50% savings over ASHRAE 90.1-2004 when all requirements in all categories are met. Have the mechanical engineer review the applicable AEDG requirements for the project type. If the project is expected to have unique systems, potential equipment is not listed, or the

system capacity is not likely to fall within the ranges in the AEDG, then the project team cannot pursue Option 2, and must pursue Option 1, Option 3, or (for Data Centers) Option 4.

Option 3. Systems Optimization savings vary dependent on the efficiency strategies, climate zone and building type.

Energy Modeler Qualifications

Refer to the LEED v4 reference guide.

Developing a Preliminary Energy Model

Refer to the LEED v4 reference guide, with the following addition:

- ▶ See ASHRAE 209 Sections 6.3 - Load Reduction Modeling for further guidance.

Modeling HVAC Systems

Refer to the LEED v4 reference guide, with the following addition:

- ▶ See ASHRAE 209 Section 6.4 - HVAC System Selection Modeling for further guidance.

Building the Proposed Building Performance Model

A team that has already prepared a preliminary model may update it to reflect the newest design information throughout the project.

Create or update proposed building characteristics based on the latest information and specifications on systems, assemblies, and equipment in the current design. This can be accomplished as early as design development to estimate projected savings, and later updated when the construction documents are complete. Then analyze remaining efficiency strategies that the team would like to consider before the design documents are finalized. For example, the proposed building performance energy model could be used to evaluate the performance and cost implications of value engineering decisions.

In most cases, the ASHRAE 90.1 proposed building performance model will exactly mirror the project design. However, ASHRAE 90.1 Appendix G indicates some specific cases where the modeled parameters may vary from the actual design. Examples include:

1. All conditioned spaces in the proposed design, with the exception of a few space types must be simulated as being both heated and cooled even if a heating or cooling system is not installed (Table G3.1(Proposed)(1)(b)).
2. HVAC fans used for ventilation shall be cycled on and off to meet heating loads during unoccupied hours, even if the systems are scheduled to remain off during unoccupied hours in the project design (Table G3.4(Proposed)(4)).
3. Lighting in unfinished spaces shall be modeled as meeting ASHRAE 90.1-2016 Table 9.5.1 prescriptive requirements.

Building the Baseline Building Performance Model

Developing the baseline building performance model is a detailed process that requires a good working knowledge of ASHRAE 90.1-2016, Appendix G. The baseline model represents a typical design for a building of the same size, function, and number of floors as the proposed building. It meets the prescriptive and mandatory requirements of ASHRAE 90.1-2004 for a building with standard practice HVAC, lighting, plumbing and envelope systems.

In general, baseline building performance energy model development begins by changing the inputs for all the components, assemblies, systems, and controls of the proposed building performance energy model to values, types, and controls prescribed in accordance with 90.1-2016 Appendix G. Whereas previous versions of ASHRAE 90.1 Appendix G required the energy modeler to determine Baseline parameters by referring to the prescriptive requirements in Sections 5 through 10 of the standard, ASHRAE 90.1-2016 Appendix G is self-contained, and includes the relevant referenced requirements within the Appendix. This should simplify the Baseline modeling process for projects using Appendix G.

Determine or update all relevant baseline inputs for the appropriate climate zone, building type, and building area.

When developing the baseline building performance model, ensure that additional HVAC system types in addition to the predominant HVAC system type are modeled as required in G3.1.1 (b) through (h). Spaces that are served by a different HVAC system in the proposed design due to load or schedule variances, different building functions, or cross-contamination requirements, will also often be served by a different system type in the baseline building due to the requirements stipulated in G3.1.1(b) through (h). Examples include:

- ▶ Per G3.1.1(b), a security office operating 24x7 and a kitchen with high peak summer cooling loads located in a midrise office building will each be modeled with a single zone system in the Baseline due to peak thermal loads that differ by 10 Btu/h*² or more from the average of other spaces served by the system, and/or schedules that differ by more than 40 equivalent full-load hours from other spaces served by the system.
- ▶ Laboratory spaces in buildings with significant laboratory exhaust will be modeled as a single VAV system serving only those spaces per G3.1.1(d).
- ▶ A mixed use residential and non-residential building will be modeled with both residential and non-residential system types if the total area associated with each space type exceeds 20,000 square feet per G3.1.1(b).
- ▶ A heated only warehouse space will be modeled with a heated-only system (System type 9 or 10) while the adjacent office area will be modeled with both cooling and heating per G3.1.1(e).

If the energy simulation software automates some or all the baseline generation, review the automated baseline model inputs against the expected baseline values and confirm consistency (see *Further Explanation, Common Issues with Energy Modeling*).

Preparation of the initial baseline building performance model is best undertaken during the design development phase, after major design decisions have been made, so that modeling can evaluate whether the project is likely to meet energy savings targets (or achieve points under the related credit). The baseline building performance model will typically need to be updated upon completion of the final project design.

Schedules

If anticipated operating schedules are unknown, helpful guidance for determining model inputs for occupancy, lighting, HVAC system, receptacle power, and service hot water consumption values can be found in the ASHRAE 90.1-2016 User's Manual, Appendix G.

Align the time steps used in the modeled schedules with the time steps used for determining peak demand. For elevators, the modeled peak power for the elevator motors must either be scaled down from the instantaneous peak by an equal factor in the baseline and proposed design to represent the hourly peak, or the fractional schedule modeled for elevators shall have a peak operating percentage that aligns with the peak hourly demand as a fraction of the peak instantaneous demand. For example, if the peak elevator motor power for the proposed design is 100 kW, the peak elevator motor power for the baseline design is 120 kW, and the peak hourly demand for the proposed elevators is 5 kW, the elevator motor power shall either be modeled as: hourly schedules that peak at 100% of design power, and peak hourly demand modeled as 5 kW in the Proposed Design and 6 kW in the Baseline design; or hourly schedules that peak at 5% (including for Baseline HVAC system sizing), and peak hourly demand modeled as 100 kW in the Proposed Design and 120 kW in the Baseline design.

Similar schedule or peak hourly load adjustments apply for receptacle power (with maximum connected load versus peak hourly demand) and service water heating (with maximum instantaneous flow versus peak hourly demand).

Schedules must be identical in both the baseline and the proposed cases unless documented in an exceptional calculation or specifically allowed by ASHRAE 90.1-2016 Appendix G (*see Further Explanation, Exceptional Calculation Method*).

Certain space types may require specific schedules based on anticipated operation and may vary by space type. For example, a server room may have different temperature schedules than an occupied space. Exceptions to Section G3.1.1 may require modeling of a different baseline HVAC system type in spaces with schedules that vary significantly from the rest of the building.

Different lighting schedules should be used for a project with both office and retail occupancy when the space-by-space method is used, or when the building area method is used with multiple building type classifications. Different schedules cannot be used, however, if an average lighting power density is applied to the whole project.

Energy Cost

For EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance, baseline and proposed costs must be determined on an energy cost basis using actual utility rates or the state's average energy prices referencing US Energy Information Administration (EIA) state average annual rates per fuel source for the most recent year for projects in the U.S., or the most recent published national or provincial average annual rates per fuel source for projects outside the U.S.

For each energy source serving the building, the utility rates or tariffs must be identical for the Baseline and Proposed building models.

For projects using actual rates:

- ▶ The modeled rates shall include all components of the rates applicable to the building (i.e. consumption charges, demand charges, monthly or time-of-use variations in rates, distribution and transportation charges, etc.). Monthly or average annual rates shall be limited to applications where this is the only information available to the project (i.e. campus projects where blended rates are calculated for the whole campus).
- ▶ If the baseline model has a fuel source not included in the proposed design (i.e. natural gas or propane), the rates shall be determined based on current rates available in the project location (e.g. a current natural gas tariff applicable to similar buildings that have natural gas service in the project location).

Energy cost savings is referenced as a metric for overall building energy efficiency:

- ▶ It aligns with the energy modeling procedures in ASHRAE 90.1-2016, Appendix G
- ▶ It captures the relative effects of various efficiency measures on energy demand and long-term operating costs—valuable metrics for the owner in determining the overall cost-effectiveness of selected efficiency strategies.
- ▶ It can help designers understand energy consumption because in many cases, cost and environmental impacts of each fuel source are correlated.

For the cost metric, on-site renewable energy that complies with ASHRAE 90.1-2016 Section G2.4.1 requirements may be documented for credit under EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Off-site renewable energy and any on-site renewable energy not meeting the criteria of Section G2.4.1 is ineligible for credit when applying the cost metric (See [Further Explanation, Applying Renewable Energy Savings](#)).

Greenhouse Gas Emissions

LEED v4.1 incorporates greenhouse gas emissions (also referred to as CO2 equivalent emissions) as a metric for building energy performance. Understanding greenhouse gas emissions from building energy use and prioritizing building emissions reductions is critical for addressing climate change.

The total greenhouse gas emissions metric, in terms of carbon dioxide equivalents, shall be calculated for the baseline building performance rating and for the proposed building performance rating, and the percentage improvement shall be determined using carbon dioxide equivalent emissions. For each

energy source serving the building, the GHG emission factors must be identical for the Baseline and Proposed building models.

Greenhouse gas emissions factors per fuel source shall be determined as follows (the most recently available annual data shall be used unless otherwise indicated):

- ▶ United States or Canada:
 - Use U.S. Environmental Protection Agency's (EPA) Greenhouse Gas Factors for the most recently available year (see Energy Star Portfolio Manager Technical Reference: Greenhouse Gas Emissions or <https://www.epa.gov/eGRID/data-explorer>).
 - For electricity, use indirect greenhouse gas emission factors per eGRID region in the U.S., and indirect greenhouse gas emission factors per province in Canada.
 - For fuel (including biomass), use direct GHG emissions factors per fuel source, as applicable.
 - For purchased heating or purchased chilled water, use indirect GHG emissions factors per fuel source.

OR

- Continental US (required if documenting Optimize Energy Performance Credit greenhouse gas emissions savings for Tier 2 new off-site renewables)
 - Electricity hourly emissions factors: Use hourly Cambium Levelized Long-Run Marginal Emission Rates published by NREL for the project's Generation and Emission Assessment (GEA) region with characteristics as defined in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*. The hourly electricity consumption is multiplied by the hourly CO₂e emissions factor to determine the hourly CO₂e emissions.
 - Other Fuels: Use the total CO₂e emissions factors that include both direct and indirect CO₂e referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

Note: These references account for both indirect and direct emissions (i.e. combustion + pre-combustion), leading to higher reported GHG emissions for fossil fuel and for the portion of eGRID electricity generated by fossil fuel).

- ▶ European Union:
 - European Environment Agency (EEA) – National emissions reported to the United Nations Framework Convention on Climate Change (UNFCCC) and complete energy balances from Eurostat. (<https://www.eea.europa.eu>).
- ▶ All other locations, and EU countries without EEA data reporting
 - Climate Transparency - Enerdata national emissions factors (<https://www.climate-transparency.org>; <https://www.enerdata.net/>).

For references above that report only electricity emissions factors, use the U.S. Environmental Protection Agency's (EPA) direct GHG emissions per fuel source published in the Energy Star Portfolio Manager Technical Reference: Greenhouse Gas Emissions for all non-electric fuel sources.

Refer to *Further Explanation, Applying renewable savings* for further guidance on demonstrating renewable energy savings within the energy model. For projects in the continental United States, note that the contract for Tier 2 New Off-site renewable energy must include sufficient information to identify the grid region, and a reasonable estimation of the hourly generation profile to contribute to GHG emissions savings in EAc Optimize Energy Performance.

Custom GHG Emissions factors:

Projects outside the U.S. may document custom GHG Emissions factors to be approved on a case-by-case basis for published national or regional CO₂ equivalent emissions per fuel source, particularly references that include hourly electric emissions factors. Supplemental documentation shall be provided to demonstrate that the methodology for calculating the emissions factors, and assumptions per fuel input source are derived consistently. For reporting methods that separate “renewable” and “non-renewable” emissions (e.g. for wood fuel classified as renewable in the project location; or for national average electric emissions that separately report renewable versus non-renewable emissions), the total renewable plus non-renewable emissions factors shall be used. For non-electric fuel sources, if the electricity emissions factors are calculated based on fuel inputs that include both indirect and direct emissions (i.e. combustion + pre-combustion), use the total CO₂e emissions factors that include both direct and indirect CO₂e referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

Electricity hourly emissions factors:

For a more granular picture of greenhouse gas emissions reduced through building efficiency, renewables procurement, and grid harmonization strategies, projects are encouraged to use hourly electricity emissions profiles when available. In the continental United States, NREL has published Cambium hourly long run marginal emissions profiles estimated based on prospective power sector scenarios for each Generation and Emission Assessment (GEA) region. The specific power sector scenario and characteristics used for LEED documentation are referenced in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator*.

Hourly emissions for baseline electricity, baseline unregulated electricity, and proposed electricity are calculated by multiplying the hourly electricity greenhouse gas emission factors from the project’s grid region by the hourly electric consumption from the energy simulation for the baseline design, baseline design unregulated electricity, and proposed design respectively. The hourly electric emissions for each case are summed together to determine the associated annual greenhouse gas emissions. Hourly emissions for qualifying Tier 2 off-site renewable energy are determined based on the renewable asset’s GEA grid region and hourly electricity generation from the renewable asset (See Further Explanation, Applying renewable savings).

Some energy software can include this hourly emissions calculation directly into the energy model (similar to a utility rate), while other energy software may require post-processing of hourly electric consumption to perform the simple calculation. For projects in the continental United States, the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator* completes this post-processing based on hourly electricity data and monthly non-electric energy data imported into the calculator from the energy model. Most energy simulation software provides hourly and monthly energy outputs in a delimited format and can be quickly imported into the calculator.

For prerequisite compliance, only renewable energy meeting the ASHRAE 90.1-2016 definition for on-site renewable energy may be modeled for credit when using the GHG metric for compliance. For Optimize Energy Performance credit compliance, Tier 2 off-site renewable energy may also be modeled for credit. For projects claiming credit for GHG emissions reductions associated with Tier 2 off-site renewable energy in EA credit Optimize Energy Performance, building GHG emissions and avoided GHG emissions from Tier 2 off-site renewable energy can be modeled using the hourly generation profile from the project’s grid region and the renewable generation grid region in locations where hourly emissions factors are available.

Applying Renewable Energy Savings

Credit for on-site and new off-site renewable energy may be documented as follows, either directly in the proposed energy model, or applying post-processing of the proposed energy model results.

► **On-site renewable energy:**

Energy costs and greenhouse gas emissions savings associated with on-site or on-campus renewable energy systems may be documented for EA Prerequisite Minimum Energy and EA Credit Optimize Energy Performance when the following criteria are met:

- Per ASHRAE 90.1-2016 Section G2.4.1, the renewable systems are included on the building permit and used in the building. LEED also recognizes renewable systems that are on a master site permit for a contiguous campus that includes the LEED project when the renewable energy is used in the contiguous campus.
- Energy is generated from renewable sources produced at the building site or master site per ASHRAE 90.1-2016 definitions.
- All associated environmental attributes are retained by the building owner.

Note: earlier versions of LEED allowed some biofuels produced off-site to qualify as on-site renewable energy. However, based on the clarifications provided in ASHRAE 90.1-2016 for on-site renewable energy, and the clearer distinction between on-site and off-site renewable energy in LEED v4.1, biofuels are only considered on-site renewable systems when the renewable source is harvested on site or on a contiguous campus, and used for on-site generation of electric or thermal energy. Furthermore, the renewable system must be part of the LEED project scope of work (or campus development scope of work including the project) under the ASHRAE 90.1-2016 requirements. Therefore, electric generation or thermal generation from renewable fuels sourced off-site and transported to the building site are not considered on-site renewable generation.

The renewable energy generation shall be documented using a method that accounts for all losses, with relevant design criteria and local climatic conditions incorporated into the calculations. Free web-based tools produced by U.S. national laboratories are available for estimating annual, monthly, and hourly energy generation from solar photovoltaics. For on-site renewable generation of electricity, projects using net metering may claim credit for all electric generation on an annual basis, up to 100% of the electricity documented in the energy model (including electricity associated with District Energy Systems serving the project, and documented using v4 Reference Guide, *Further Explanation, Project Type Variations, District Energy Systems, Full DES Performance Accounting*). For on-site renewable generation of thermal energy, documentation shall be provided to support that the hourly thermal energy generation, available thermal storage capacity, and relevant thermal losses are evaluated in conjunction with the hourly thermal load profile.

ASHRAE 90.1-2016 Section G2.4 states that on-site renewable energy “shall be subtracted from the *proposed design energy* consumption prior to calculating the *proposed building performance*.”

- Cost Metric: The equivalent cost of the on-site renewable energy system can be calculated in two ways, virtual rate or actual utility tariff.
 - Virtual rate. The project team may use the virtual energy rate determined by the proposed building energy model used for EA Credit Optimize Energy Performance. The virtual rate accounts for both consumption and demand charges. Project teams that use the Energy Information Administration’s average energy prices must use the virtual rates to determine the renewable energy system cost.
 - Actual utility tariff. Calculate the expected savings in both consumption and demand charges, based on the rates charged by the utility that serves the project. If a project is served by a utility that uses time-dependent valuation to set rates, the team must provide hourly calculations for the value of generated energy. Some energy modeling software may calculate the savings from renewable energy systems if the utility rates include consumption, demand, time-dependent valuation, time-of-use, ratchets, and other factors.

Annual cost savings for renewable electricity generation shall not exceed 100% of the annual electricity cost documented in the proposed energy model before applying renewable savings.

- Greenhouse gas emissions metric: the proposed emissions associated with the on-site renewable generation shall be calculated using the same methodology as the proposed design (i.e. annual average GHG emissions factors per energy source, or hourly electric emissions). If using hourly electric emissions, the project must document the hourly renewable generation profiles for any on-site renewable electric generation. Annual GHG emissions savings for on-site renewable electric generation on an annual basis shall not exceed 100% of the annual electric GHG emissions documented in the energy model before applying renewable savings.

► **Tier 2 New Off-site renewable energy:**

Greenhouse gas emissions offset by Tier 2 New off-site renewable energy systems qualifying under EA credit Renewable Energy may be included in the model for achievement of points using the greenhouse gas emissions calculation under EA credit Optimize Energy Performance, but may not be included in the model for EA Prerequisite Minimum Energy Performance compliance.

Hourly GHG Emission requirements:

Projects located in the continental United States or in any other location where hourly GHG electric greenhouse gas emissions factors are available must document performance for the baseline, proposed design, and renewable generation using hourly electric emissions factors. Therefore, for projects in the continental United States, the contract for the renewable asset must include sufficient information to identify the grid region, and a reasonable estimation of the hourly generation profile for the renewable asset. Many larger contracts include the renewable project location and renewable generation hourly profile in the contract documents, as a set of twelve monthly profiles, indicating the typical 24-hour generation profile for each month of the year. For smaller contracts, if the renewable asset location and characteristics are indicated on the contract, the hourly generation profile can be estimated:

- Photovoltaic generation: If the photovoltaic generation location, module type, array type, system losses, tilt, and azimuth are known, NREL's free online PVWatts tool may be used to calculate the hourly solar generation profile.
- Wind generation: If the wind generation location and annual generation amount are known, hourly wind generation data from the US Hourly electricity grid monitor for the balancing authority closest to the project generator may be used to generate an hourly profile, calculating the hourly wind generation as a percentage of total annual wind generation for that location (<https://www.eia.gov/electricity/gridmonitor/>).

Avoided hourly GHG emissions from Tier 2 New off-site renewable energy are calculated by multiplying the hourly electricity greenhouse gas emission factors from the renewable asset's grid region by the hourly electric generation from the renewable energy system. For projects in the continental United States, these hourly calculations are performed automatically in the *LEED Hourly Cambium, Demand Adjusted Energy Metrics Calculator* based on data inputs indicating the hourly renewable generation and the renewable asset's grid region. Annual GHG emissions savings for renewable electric generation shall not exceed 100% of the annual electric GHG emissions documented in the proposed energy model before applying renewable savings.

- Tier 1 on-site renewable energy systems that do not meet the contractual requirements of ASHRAE 90.1-2016 Section G2.4 may document achievement of points using the greenhouse

gas emissions calculation under EA credit Optimize Energy Performance using the guidance for Tier 2 New off-site renewable energy systems.

Exceptional Calculation Method

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Replace all instances of ASHRAE 90.1-2010 with ASHRAE 90.1-2016
- ▶ Delete the first paragraph in the *Additional guidance* section
- ▶ Delete the *Changes from earlier versions of ASHRAE and LEED* section
- ▶ Refer to Rating System Variations, Data Centers for exceptional calculations that may be applied to data centers that comprise a portion of the project area.
- ▶ Refer to Rating System Variations, Retail for exceptional calculations that may be applied to commercial cooking equipment, appliances, or commercial refrigeration.

ASHRAE 90.1-2016 versus 2010

The referenced standard for building the baseline model for this prerequisite has been updated to ASHRAE 90.1-2016, which represents a substantial increase in efficiency from ASHRAE 90.1-2010. Some of the major changes are described in Tables 1 and 2. Detailed changes between ASHRAE 90.1-2010 and ASHRAE 90.1-2013 are described in ASHRAE 90.1-2013 Appendix F. Detailed changes between ASHRAE 90.1-2013 and 90.1-2016 are summarized in ASHRAE 90.1-2016 Informative Appendix H.

Table 1. Changes in ASHRAE 90.1 mandatory requirements, 2010 to 2016

Building Envelope Requirement	ASHRAE 90.1-2016
Heated or Cooled Vestibule requirement (6.4.3.9)	Requires heated or cooled vestibules to limit setpoint temperatures, and automatically shut off heating when outdoor air temperature exceeds a certain level.
Verification of Envelope requirements (4.2.4, 4.2.5, 5.2.1, 5.2.9)	Adds verification requirements for envelope components including insulation, air leakage, and other properties.
HVAC & Refrigeration Requirement	ASHRAE 90.1-2016
Refrigerators and Freezers (Tables 6.8.1-12 and 6.8.1-13)	Maximum energy consumption regulated for some commercial refrigerators and freezers
HVAC Equipment (Tables 6.8)	Increased efficiencies for HVAC equipment, and increased capacity control for some packaged equipment
Humidification and Dehumidification (6.4.3.6)	Increased control requirements prohibiting the use of fossil fuel and electricity for humidification above 30% RH and dehumidification below 60% RH in most circumstances.
Demand Control Ventilation (6.4.3.8)	Reduces the occupancy threshold where DCV is required from 40 people per 1,000 ft ² to 25 people per 1,000 ft ²
Heating and Cooling Setbacks (6.4.3.3)	Requires heating setback at least 10°F (6°C) below occupied heating setpoint, and cooling setback at least 5°F (3°C) above occupied cooling setback. <i>Note: These setbacks must be part of the Baseline and Proposed schedules modeled using the Performance method.</i>
Optimum start control (6.4.3.3.3)	Optimum start controls required for more building types
Duct Insulation (6.4.4.1.2)	Increases ductwork insulation requirements

DDC control (6.4.3.10)	DDC Control required for a much larger array of building applications
HVAC alterations (6.1.1.3.1)	Requires replacement HVAC&R equipment to meet most requirements
Pool Dehumidifier (6.4.1.1)	Establishes efficiency requirements for indoor pool dehumidifier
Fault Detection (6.4.3, 6.4.3.12)	Adds fault detection requirements
Power Requirement	ASHRAE 90.1-2016
Automated receptacle control (8.4.2)	Expands the spaces where automated receptacle control is required and provides further details regarding acceptable methods for receptacle controls
Electrical Monitoring (8.4.3)	Adds monitoring requirements to submeter tenant energy and electric end uses
Automated receptacle control (8.4.4)	Adds transformer performance requirements
Lighting Requirement	ASHRAE 90.1-2016
Daylighting Controls (9.4.1.1)	Requirements updated for areas where mandatory daylighting controls are required.
Automatic shutoff of lighting and switched receptacles in hotel guestrooms (9.4.1.3)	Adds requirements for automated shutoff of lights and switched receptacles in hotel/motel guestrooms
Lighting Controls (9.4.1.1)	Additional lighting controls requirements including partial automatic ON, inclusion of emergency circuits in scheduled shutoff requirements, additional shutoff controls for exterior lighting, increased parking garage occupancy controls.
Lighting Efficacy (9.4.1)	Adds efficacy requirements for residential dwelling unit lighting
Lighting Alterations (9.1.2)	Increases requirements for alterations to existing building lighting systems
Exterior Lighting Power (9.4.2)	Reduces exterior lighting power allowances
Motor Requirement	ASHRAE 90.1-2016
Motor Efficiency (10.4.1)	Increases motor efficiencies
Escalators (10.4.3 and 10.4.4)	Adds requirements for escalators, moving walkways, and elevators
Whole Building energy monitoring (10.4.5)	Adds requirement to monitor whole building energy use for energy supplied by a utility, energy provider, or plant not located in the building

Table 2. Changes in ASHRAE 90.1 prescriptive requirements, 2010 to 2016

Building Envelope Requirement	ASHRAE 90.1-2016
Opaque and Fenestration Efficiencies (Tables 5.5-1 through 5.5-8)	More stringent insulation levels for opaque elements in most climate zones. Fenestration: More stringent U-factor requirements for most assemblies, more stringent SHGCs in warmer climates. Additional fenestration framing types added.
Fenestration area by orientation (5.5.4.5)	Specific limitations added for fenestration area by orientation

Fenestration Visible Transmittance (5.5.4.6)	Minimum visible transmittance to solar heat gain coefficient ratio added.
HVAC & Refrigeration Requirement	ASHRAE 90.1-2016
Heat Rejection Fan Control (6.5.5.2)	Fan control required for multi-cell heat rejection equipment
Cooling tower flow turndown (6.5.5.4)	Cooling towers with multiple or variable speed condenser water pumps have added controls requirements associated with flow rate
Small motors (6.5.3.5)	Most motors under 1 hp required to be electrically commutated or have minimum efficiency of 70%.
Boiler Turndown (6.5.4.6)	Large boilers required to have minimum turndown ratio
Fan Power Allowance (Table 6.5.3.1B)	Changes to fan power pressure adjustments. Some allowances previously allowed to be used for a broad range of systems such as fully ducted return and exhaust are limited to specific systems.
Dehumidification (6.5.2.3)	Requires most reheat used for dehumidification to be from recovered or site-generated sources
Fluid Flow (6.5.4.1 through 6.5.4.3)	Requires automatic shutoff of pumps and boilers when fluid flow through the chillers or boilers is not operating, reduces low flow limit exceptions, requires variable flow in more hydronic system applications
Computer rooms (6, 6.6)	Adds requirements specific to computer rooms, including air and water economizer requirements
Transfer air (6.5.7.1)	Limits conditioned supply of transfer air between spaces
VFD Return and Relief Fans (6.5.4.1, 6.5.4.3)	Requires VFD control of return and relief fans larger than 0.5 hp.
Fan Powered VAV control	Specifies control of fans in fan-powered parallel VAV boxes
Energy Recovery (Tables 6.5.6.1-1 and 6.5.5.6.1-2)	Revises minimum threshold for energy recovery
Water-side economizers	Requires water-side economizers for radiant cooling or passive chilled beam systems
Lighting Requirement	ASHRAE 90.1-2016
Interior Lighting Power Density (Tables 9.5.1 and 9.6.1)	Extensive changes to the Interior Lighting Power Density requirements.
Decorative Lighting (9.6.2)	Reduces additional lighting allowance for decorative lighting

Table 3. Changes in ASHRAE 90.1 Performance Rating Method Requirements, 2010 to 2016

General Requirement	ASHRAE 90.1-2016 Appendix G
Performance Rating Method Scope (G1.1)	Appendix G can be used to demonstrate code compliance (Previously it only applied to demonstrate above-code performance).
Performance Rating Calculation (G1.2.2 / 4.2.1.1)	A stable baseline that references ASHRAE 90.1-2004 prescriptive values is introduced. Performance Cost Index Target (PCI_T) is calculated using Building Performance Factors (BPF) for each building type and climate zone in conjunction with Baseline Building Unregulated Energy Consumption (BBUEC) and Baseline Building Performance (BBP).

	Building Performance Factors represent the ratio of regulated energy cost for a 90.1-2016 versus a 90.1-2004 compliant building. This allows relatively few changes to the Baseline Building modeling methodology between code cycles, with the major change being the BPF determinations.
Self-contained references	The Baseline modeling requirements are contained within ASHRAE 90.1-2016 Appendix G, and do not require references to the prescriptive requirements of Sections 5 through 10.
Unmodified existing building components (Table G3.1#2(Baseline))	Unmodified existing building components are required to follow the same rules as new and modified building components. Previously some existing building components (such as existing building envelope components) could be modeled using existing unrenovated performance in the Baseline and as-designed with renovations in the Proposed.
Unfinished spaces	For unfinished spaces, the proposed efficiencies, controls, lighting power densities, etc. are modeled consistent with the ASHRAE 90.1-2016 prescriptive requirements, and are not modeled identically to the Baseline.
Schedules	ASHRAE 90.1-2016 Appendix G
HVAC setpoint schedules (Table G3.1#4)	Projects may adjust schedules to demonstrate credit for HVAC systems that automatically provide occupant thermal comfort via means other than direct control of air dry-bulb and wet-bulb temperature.
Building Envelope Requirement	ASHRAE 90.1-2016 Appendix G
Vertical Fenestration Area (Table G3.1#5(Baseline)(c))	Vertical fenestration area modeled in the Baseline is less than 40% for many building occupancies. Credit is allowed when the proposed vertical fenestration area is lower than the values shown in Table G3.1.1-1 for applicable building types.
Infiltration (Table G3.1#5(Proposed)(b), G3.1.1.4)	Specific infiltration rates are required to be modeled. Credit is allowed for improved infiltration for projects performing air leakage testing.
HVAC & Refrigeration Requirement	ASHRAE 90.1-2016 Appendix G
HVAC System Type (G3.1.1, Table G3.1#10, Table G3.1.1-3, Table G3.1.1-4)	Baseline system heating type is dependent on climate zone rather than predominant heating source used in the building. Baseline systems in climate zone 0 to 3A are modeled with electric heating and baseline systems in climate zones 3B through 8 are modeled with fossil fuel heating. Baseline fossil fuel heating systems shall always be modeled using natural gas, or propane in locations where natural gas is not available. Previous versions of Appendix G required the fuel type to be the same in the Baseline and Proposed case.

	Further clarity is provided for identifying the order of priority for determining the Baseline HVAC system types applicable for the building. Additional HVAC system type categorizations added for public assembly, retail buildings up to two floors, hospitals, computer rooms.
HVAC equipment efficiencies (G3.1.2.1)	Projects are required to model both part load and full load efficiencies per Tables G3.5.1 through G3.5.6 where applicable. A clear method is provided for calculating the modeled Baseline cooling and heating COP for packaged equipment.
Night-time fan cycling (G3.1.2.4)	For System 6 and 8 (Parallel fan-powered VAV terminals with electric heating), the terminal unit fan and reheat coil are energized to meet the heating unoccupied setpoint in the space rather than the entire VAV system serving the floor.
Computer room fluid economizer (G3.1.2.6.1)	Computer room fluid economizers required for computer rooms where the Baseline system type is system 11.
Baseline Humidity Controls (G3.1.3.18 and Table G3.1#10(Baseline))	If the Baseline system type does not comply with humidistatic control requirements, then only 25% of system reheat energy shall be included in the baseline building performance. If the proposed design includes humidification, the baseline design shall use adiabatic humidification
Baseline Preheat (G3.1.3.19)	Preheat is required to be modeled for Baseline Systems 5 through 8, and controlled to a fixed setpoint 20°F (11°C) less than the design room heating temperature set point. Modeling of preheat in the Baseline is no longer dependent on the presence of preheat in the Proposed design.
Baseline Refrigeration (Table G3.1#17)	Refrigeration equipment is required to be modeled as specified.
Lighting Requirement	ASHRAE 90.1-2016 Appendix G
LPD Modeling Method (Table G3.1#6(Baseline))	All building spaces are modeled using the Space-by-Space Method. (Previous versions of ASHRAE allowed projects to use the space-by-space or building-area methods).
Automated Controls (Table G3.1#6(Proposed))	Control credit is modeled in the Proposed design for all spaces where applicable controls are included (including mandatory controls required by 90.1-2016).
Service Water Heating	ASHRAE 90.1-2016 Appendix G
Baseline System Type (Table G3.1.1-2)	Baseline service water heating system type is determined based on building type, with electric resistance point-of-use for convenience store, electric resistance storage for most commercial applications with low service water heating usage, and gas storage water heaters for residential buildings and commercial applications with high service water heating usage. In previous versions of ASHRAE, the service water heating type was modeled identically in the Baseline and Proposed Case
Service Water Heating Loads (Table G3.1#11(Baseline)(h))	A specific methodology is used for determining service water heating loads. Loads must be modeled identically in the baseline and proposed case, except when calculations show savings associated

	with reduced fixture flow, reduced required temperature of service mixed water, heat recovery for makeup water, etc.
Power and Equipment Requirement	ASHRAE 90.1-2016
Computer room equipment schedule (G3.1.3.16)	The computer room equipment schedule is varied monthly between 25% and 100% of full load as noted.
Elevators (G3.9.2, Table G3.1#16)	A specific methodology is provided for calculating baseline and proposed annual elevator energy consumption, Baseline elevator peak motor power, baseline elevator cab ventilation, and baseline elevator lighting power density.

Additional Energy Modeling Guidance

Thoroughly review both ASHRAE 90.1-2016 and the 90.1-2016 User's Manual. The manual presents extended explanations and also includes examples of the concepts and requirements within the standard.

The Pacific Northwest National Laboratory (PNNL) ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual also provides detailed modeling guidance which can be used when developing a 90.1-2016 Baseline and Proposed model

(https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-26917.pdf).

Rating System Variations

Core and Shell

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Replace instances of ASHRAE 90.1-2010 with ASHRAE 90.1-2016
- ▶ Energy cost savings are based on a building's total annual energy consumption, rather than on the owner's scope of work, so the owner of a core and shell project may have only a limited opportunity to improve energy savings. The percentage improvement thresholds for achievement of Option 1 in EA credit Optimize Energy Performance are therefore lower for Core and Shell projects than for New Construction.
- ▶ For unfinished spaces, the proposed efficiencies, controls, lighting power densities, etc. are modeled consistent with the ASHRAE 90.1-2016 prescriptive requirements, and are not modeled identically to the Baseline.

Retail

For all process loads, define a clear baseline for comparison with the proposed improvements. the baseline and design as follows:

- ▶ Commercial kitchen equipment addressed by California EnergyWise energy cost calculators may use the energy savings and percentage annual cost savings documented in these calculators to determine the Baseline annual energy consumption and Proposed annual energy consumption associated with the equipment, without additional documentation. Commercial kitchen equipment that has California EnergyWise options for natural gas or electricity may optionally be modeled using natural gas in the baseline and electricity in the proposed design, or may be modeled using the same energy source in the baseline and proposed.
- ▶ *Appliances and equipment.* For appliances and equipment not covered by California EnergyWise energy cost calculators, indicate hourly energy use for proposed and budget equipment, along with estimated daily use hours. Use the total estimated appliance/equipment energy use in the energy simulation model as a plug load. Reduced use time (schedule change) is not a category of energy improvement in this credit. ENERGY STAR ratings and evaluations are a valid basis for performing this calculation.
- ▶ *Display lighting.* For display lighting, use the space-by-space method of determining allowed lighting power under ANSI/ASHRAE/IESNA Standard 90.1-2016, with errata (or a USGBC-

approved equivalent standard for projects outside the U.S.), to determine the appropriate baseline for both the general building space and the display lighting.

- ▶ *Refrigeration.* For hard-wired refrigeration loads, model the effect of energy performance improvements with a simulation program designed to account for refrigeration equipment.

Data Centers

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Replace the *Modeling requirements* section with the section below.
- ▶ The computer room equipment schedule is varied monthly between 25% and 100% of full load as noted in ASHRAE 90.1-2016.

Modeling requirements

Energy modeling is required for all data center projects.

IT equipment energy and electrical infrastructure energy savings

Because of the high process loads associated with IT equipment and its electrical infrastructure, many project teams look to these traditionally unregulated energy end uses for energy savings. Though not required, if the project team is attempting to claim energy savings from these end uses, the data center calculator may provide a simplified method (see *Data Center Calculator*, below).

The reduced energy consumption of the IT and electrical equipment can help reduce HVAC energy usage. Project teams have the option of claiming the process load savings in isolation or creating an additional energy model based on the adjusted loads to capture the associated HVAC energy savings.

To determine total energy cost savings, it may be necessary to create three energy models. Below is a list of the models that may need to be created. The specific requirements of each model are detailed below.

1. Proposed model with as-designed IT loading (normal performance rating method, PRM, model)
2. ASHRAE baseline model with as-designed IT loading (normal PRM model)
3. ASHRAE baseline model with “baseline” IT loading (optional)

If the project team is claiming energy savings related to the IT systems, the total energy savings are calculated between models 1 and 3.

Proposed model with as-designed IT loading (model 1)

The model of the building’s energy cost must include all regulated energy end uses as listed in the prerequisite criteria, as well as any unregulated energy that is building-specific. The proposed design must use the IT loads and developed for the project and the schedule stipulated in ASHRAE 90.1-2016. The IT loads should be at the values for the intended final buildout of the facility. All electrical system components—examples include incoming transformers, switchgear, UPS systems, and power distribution units—must be modeled. Power losses associated with this equipment should be assigned to the spaces that house the equipment as an electrical load and as a thermal load input to the energy model. Model the quantity of power and cooling equipment designed to run during normal operation to include the effects of operating redundant equipment at partial loading on energy use.

In addition to the ASHRAE 90.1 mandatory compliance requirements, provide energy efficiency data for the following items:

- ▶ Generator block heaters (wattage required to keep the block at the design temperature)
- ▶ Power distribution wiring
- ▶ Battery charging

Submit documentation for the following items, showing efficiency data at initial and full system loading points (loading values are a percentage of total IT load):

- ▶ Service transformers
- ▶ Switchgear

Uninterruptible power systems

► Power distribution units

ASHRAE model with full IT loading (model 2)

Model using the same IT load as the Proposed design, with Baseline inputs consistent with ASHRAE 90.1-2016 Performance Rating Method requirements.

ASHRAE Baseline model with “baseline” IT loading (model 3)

This model is used to calculate IT energy savings due to low-energy servers, virtualization, and efficient electrical system design. In contrast to the standard application of exceptional calculation methods to the proposed model, for data center projects, the exceptional calculation is applied to the baseline (model 3). Rather than reducing the energy used in the proposed design, the baseline is increased to reflect the energy usage typical of a data center.

For IT equipment, the USGBC data center calculator provides baseline documentation; if used, additional justification for the baseline IT loads is not necessary. IT equipment input is defined as the IT load as measured at the point of connection of the IT device to the electrical power system. IT equipment input captures the actual power load of the IT device exclusive of any power distribution losses and loads beyond IT devices, such as rack-mounted fans.

The losses associated with all UPS equipment, including that which serves mechanical equipment to achieve continuous cooling during a loss of power (e.g., pumps, air-handling units, and compressors), is considered not part of the IT energy usage but part of the energy consumption required to operate the data center.

If a hydronic cooling system is used for IT cabinets or computers, the energy consumed by the fans built into the cabinet and coolant distribution pumps should be considered HVAC energy use, not IT energy use.

Data Center Calculator

Refer to the LEED v4 reference guide.

Project Type Variations

District Energy Systems

Projects that are served by district energy systems (DES) may demonstrate compliance with EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance by following one of the following methods.

- Path 1 ASHRAE 90.1-2016 Appendix G. No credit is documented for the purchased energy systems.
 - Path 1A. ASHRAE 90.1-2016 Appendix G.
 - Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a. (Revise the Appendix G methodology to remove the inherent penalty for DES)
- Path 2 Full DES performance accounting. Credit is documented for the purchased energy systems. The proposed design is modeled using a virtual plant consistent with the district energy system performance, and the baseline design is modeled with on-site systems from ASHRAE 90.1-2016 Appendix G for site generated thermal energy.
- Path 3 Large-scale District Energy Systems. GHG emissions savings associated with the upstream system are documented using the Large-Scale DES Calculator. No energy efficiency savings (using the cost or source energy metric) are documented for the upstream system.

The modeling path chosen by the project team may depend on the relative efficiency of the DES to which the project is connected, how much DES information is available, or whether an energy model

already exists for the system. Whenever possible, incorporate system and equipment performance parameters directly into the energy simulation. Potential methods include developing efficiency curves and scheduling equipment operation and curves. Postprocessing of DES performance is acceptable if reasonable simulation methods are not available or are too onerous. All postprocessing methodologies must be fully documented.

All Paths: Scope of DES equipment inclusion

All downstream equipment must be included in the scope of EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance. Downstream equipment includes heat exchangers, steam pressure reduction stations, pumps, valves, pipes, building electrical services, and controls.

Upstream equipment is included or excluded depending on the chosen path.

Path 1A. ASHRAE 90.1-2016, Appendix G

Model the proposed and baseline designs using purchased energy according to ASHRAE 90.1-2010, Appendix G.

Published purchased energy rates or conversion factors:

Energy Cost: Per ASHRAE 90.1, model the purchased energy rates for each district energy source (purchased hot water, purchased steam, or purchased chilled water) identically in the baseline and proposed design based on actual utility rates, if actual utility rates are available.

GHG Emissions: For the GHG emissions metric, if the published reference for electricity and fossil fuel GHG emission factors also reports emission factors for district energy (purchased hot water, purchased steam, or purchased chilled water), model these published emission factors for each district energy source identically in the baseline and proposed design.

Derivation of DES purchased energy rates or GHG emission factors when unpublished:

If purchased energy rates and/or GHG emission factors are not published for the district energy sources serving the project, derive these purchased energy rates and/or emission factors leveraging the electricity or fossil fuel data. For fossil fuel, use natural gas if the building does not receive fossil fuel and the district energy fuel source is unknown.

- District Chilled Water (CHW):
 $\text{CHWFactor} = \text{ElectricityFactor} \times 0.325$
- District Hot Water (HHW):
 $\text{HHWFactor} = \text{FossilFuelFactor} \times 1.65$
- District Steam Water (Steam):
 $\text{SteamFactor} = \text{FossilFuelFactor} \times 1.85$

Where:

- For the Cost Metric (See further guidance below for purchased energy rates)
 - $\text{CHWFactor} = \text{Chilled water purchased energy rate (\$/unit energy)}$
 - $\text{HHWFactor} = \text{Hot water purchased energy rate (\$/unit energy)}$
 - $\text{SteamFactor} = \text{Steam purchased energy rate (\$/unit energy)}$
 - $\text{ElectricityFactor} = \text{Electricity purchased energy rate (\$/unit energy)}$
 - $\text{FossilFuelFactor} = \text{Fossil Fuel purchased energy rate (\$/unit energy)}$Units of energy must be consistent throughout each equation (i.e. consistently \$/kWh or \$/kBtu)
- For the GHG Emissions Metric:
 - $\text{CHWFactor} = \text{Chilled water GHG emissions factor}$

- HHWFactor = Hot water GHG emissions factor
 - SteamFactor = Steam GHG emissions factor
 - ElectricityFactor = Electricity GHG emissions factor
 - FossilFuelFactor = Fossil Fuel GHG emissions factor
- Units of each GHG emissions factor must be consistent (in weight of CO₂eq emissions per unit of energy)

Additional guidance: Cost Metric ElectricityFactor and FossilFuelFactor.

For the cost metric, in a flat rate structure, in which the building cost per unit of electricity or building cost per unit of natural gas is the same throughout the year and there are no demand charges, then those flat rates become the ElectricityFactor and FossilFuelFactor for the project cost metric. If all energy rate structures are not flat, a preliminary run of the Option 1 baseline case energy model must first be completed to identify the virtual electric rate (ElectricityFactor) and fossil fuel rate (FossilFuelFactor) for the project.

To obtain the virtual fuel rate (FossilFuelFactor) when the connected building does not use fossil fuel but the DES central plant does, use a flat rate consistent with the central plant rates or the historic average local market rates.

Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a.

The project may apply ASHRAE 90.1-2022 Addendum a to the ASHRAE 90.1-2016 Appendix G criteria. This eliminates the inherent penalty in the ASHRAE 90.1-2016 Appendix G Performance Index Targets when modeling “purchased heat” and “purchased chilled water”. Replace all ASHRAE 90.1-2022 Addendum a references to Section 6 prescriptive criteria for the *proposed building design* with ASHRAE 90.1-2016 Section 6. For the *baseline building design* HVAC systems, the project team must exclusively reference ASHRAE 90.1-2022 with Addendum a. Free read-only versions of ASHRAE 90.1-2022 are available at <https://www.ashrae.org/technical-resources/standards-and-guidelines/read-only-versions-of-ashrae-standards>.

90.1-2022 Addendum a criteria as applied to ASHRAE 90.1-2016 Appendix G:

- Model HVAC systems for the *baseline building design* per ASHRAE 90.1-2022 Appendix G criteria as if all heating and cooling generation equipment is on-site;
- Projects with purchased heat: Model the *proposed building design* using natural gas forced draft boilers that prescriptively comply with ASHRAE 90.1-2016 Section 6 in lieu of purchased heat. The number of boilers and boiler controls shall meet the requirements of ASHRAE 90.1-2022 Section G3.2.3.2 through G3.2.3.6, without exceptions. Forced draft boiler efficiencies shall be modeled per the mandatory and prescriptive requirements of ASHRAE 90.1-2016 Section 6. Boiler systems with design input exceeding 1,000,000 Btu/h may document credit for minimum turndown ratios per Table 6.5.4.1.
- Projects with purchased chilled water: Model the *proposed building design* using water-cooled chillers that prescriptively comply with ASHRAE 90.1-2016 Section 6 in lieu of purchased chilled water.
 - Model the type and number of water-cooled electric chillers per ASHRAE 90.1-2022 Table G3.2.3.7 based on the peak coincident cooling load of baseline HVAC systems using chilled water (See 90.1-2022 Section G3.2.3.7).
 - Model the chilled water (CHW) with a design supply temperature of 44 °F (7 °C) and return temperature of 56 °F (13.3 °C) (See 90.1-2022 Section G3.2.3.8)
 - Model each chiller with separate condenser-water and primary chilled-water pumps interlocked to operate with the associated chiller per ASHRAE 90.1-2022 G3.2.3.11. Model the CHW loop as constant-flow primary and variable-flow secondary with the pump power of each loop modeled per 90.1-2022 Section G3.2.3.10, without exceptions. Model secondary loops with a pump motor demand of 30% of design wattage at 50% of design flow per the prescriptive criteria of ASHRAE 90.1-2016 Section 6.5.4.2. For

systems with total modeled chilled water capacity exceeding 300,000 Btu/h (25 kW) utilizing DDC CHW control valves, model chilled water supply temperature reset based on valve positions until one valve is wide open or setpoint limits have been reached per ASHRAE 90.1-2016 Section 6.5.4.4.

- Model heat rejection as an axial fan cooling tower with design fan power = 40.2 gpm/hp per ASHRAE 90.1-2016 Table 6.8.1-7, and with design supply temperature and leaving water temperature determined per ASHRAE 90.1-2022 G3.2.3.11. If the total fan power for the heat rejection equipment exceeds 5 hp, model the cooling tower fans with variable-speed fan controls that reduce fan motor demand to no more than 30% of design wattage at 50% of design air volume per ASHRAE 90.1-2016 Section 6.5.11.

Path 2. Full DES performance accounting

For path 2, the energy model scope accounts for both downstream equipment and upstream equipment and requires calculation of the district energy average efficiencies using engineering analysis or monitored data or a combination of both.

Energy rates (Cost Metric)

All DES electricity and fuel rates must be identical in both the baseline and the proposed cases. Use local electricity and fuel rates as they would normally apply to the building for the energy sources under consideration. If this information is not available, use representative market rates.

Exception: For District cooling or district heating plants without cogeneration or fuel cells that operate under specific and atypical electric rate structures and actively take advantage of those rates through strategies such as load management or energy storage, use the rate structures as they apply to the DES.

Greenhouse Gas Emissions Factors

See the guidance in Further Explanation, Greenhouse Gas Emissions.

Baseline building systems

For systems with thermal energy delivered from the district energy system, model the baseline case with on-site systems per ASHRAE 90.1-2022 Addendum a criteria described above.

Proposed building plant

Model the proposed case with a virtual DES-equivalent plant. Use the same efficiencies as the entire upstream DES heating, and cooling, and combined heat and power (CHP) systems, including all distribution losses and energy use.

Equipment efficiencies, distribution losses, and distribution pumping energy may be determined using any of the following methods:

- Monitored data
- Engineering analysis

Efficiencies and losses may be determined and modeled at any level of time resolution, from hourly to annual. However, the time resolution must be sufficiently granular to capture and reasonably represent any significant time- or load-dependent interactions between systems, such as thermal storage or CHP. Monitoring and analytical methods may be combined as necessary and appropriate. Monitoring data for heating, cooling, pumping, and cogeneration may be used only if the thermal loads that are monitored represent at least 90% of the load on the campus or district plant predicted after building occupancy. Whether using monitoring or an analytical method, the methodologies must be fully documented. The following specific requirements apply.

Heating and cooling plants

Efficiencies, whether determined through monitoring or analytically, must include all operational effects, such as standby, equipment cycling, partial-load operation, internal pumping, and thermal losses.

Thermal distribution losses

Use monitored data or an engineering analysis.

- Monitored data determine the distribution losses for the DES by measuring the total thermal energy leaving the plant and comparing it with the total thermal energy used by the buildings connected to the DES. Rate the plant efficiency accordingly in the energy model:

Plant efficiency (%) x [100% – distribution loss (%)]

- An engineering analysis takes into consideration all distribution losses between the DES and the building. For distribution main losses, use a prorated amount based on load. For dedicated branch losses, use the total losses of the branch that feeds the building, including heat losses and steam trap losses. Compare the total losses with the total load of the building to get a percentage distribution loss relative to load and downgrade the plant's efficiency accordingly in the energy model.

If thermal distribution losses are not measured or modeled, use the following default losses:

- Chilled water district cooling, 5%
- Hot water district heating, 10%
- Closed-loop steam systems, 15%
- Open-loop steam systems, 25%

For steam systems that are partially open and partially closed, prorate between the above 15% and 25% losses in accordance with the fraction of expected or actual condensate loss.

Pumping energy

Whether through monitored data or engineering analysis, determine pumping energy for the project by prorating the total pump energy of the DES by the ratio of the annual thermal load of the building to the total annual DES thermal load. Model the pump energy as auxiliary electrical load. Pumping energy must be determined or estimated where it applies.

District Energy Combined Heat and Power (CHP)

To model the proposed design virtual plant, first monitor or model the total electricity generation, fuel input, and heat recovery associated with the District Energy Combined Heat and Power (CHP):

- Determine annual electricity generation using one of the following methods:
 - Monitor the total annual gross electricity generation. Also monitor the total annual parasitic loads, such as the annual electricity used for cooling the intake air for a turbine. Calculate the net annual electricity generation by subtracting all parasitic loads from the annual gross electricity generated.
 - Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators. Apply measured or estimated load profiles as process loads to reflect the estimated total electric and thermal loads on the district energy CHP system. Use the total energy generated and total fuel input from this analysis. Any parasitic loads must be included in the analysis and subtracted from the annual electricity generation.
- Calculate annual fuel input using one of the following methods:
 - Monitor the total annual fuel input to the generators.

- Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators.
- Calculate waste heat recovery using one of the following methods:
 - Monitor the total waste heat recovered.
 - Model the generators in energy simulation software per Appendix G. Use peak electricity efficiencies and generator curves that match the installed generators. Model the thermal equipment served by the CHP waste heat, such as boilers and absorption chillers, using the installed equipment capacities, efficiencies, and efficiency curves, and reflecting the total heating and cooling loads on the plant as a process load. Use the energy modeling outputs to identify the total heat recovered.

For baseline CHP electricity output, follow the general procedures described in this section for the proposed case, and adjust the results as follows depending on the results of the DES electricity allocation and the total modeled electricity use of the building in the Path 2 proposed case, including the electricity consumption of district plant equipment serving the building:

- Scenario A. If the building's allocation of CHP-generated electricity is less than or equal to its modeled electricity consumption, no adjustment is necessary. The baseline building is charged with the energy used by its (non-CHP) systems at market rates using standard procedures.
- Scenario B. If the building's allocation of CHP-generated electricity exceeds its modeled electricity consumption, include the amount of excess CHP electricity case as described in CHP fuel input formulas.

For the proposed design's CHP electricity output, allocate the electricity generation to the building based on the fraction of thermal loads to the building for the DES sources that use recovered waste heat. For each DES source supplied to the building, determine the fraction of the recovered waste heat applied to that source as well as the amount serving the project building. For relatively simple DES systems, in which the recovered waste heat is used directly in the DES, and for which waste heat serves only heating loads in the connected buildings, use the formula for simple systems:

$$\text{CHP_ELEC}_{\text{BLDG}}(\text{simple systems}) = (X_{\text{HEAT}} \times \text{BLDG}_{\text{HEAT}}) \times \text{CHP_ELEC}_{\text{TOTAL}}$$

where

$\text{CHP_ELEC}_{\text{BLDG}}$ = CHP electricity generation allocated to building

X_{HEAT} = fraction of CHP plant's total production of waste heat applied to the DES directly

$\text{BLDG}_{\text{HEAT}}$ = fraction of total district heat provided to building

$\text{CHP_ELEC}_{\text{TOTAL}}$ = total CHP electricity generated at DES plant

For CHP plants in which a portion of the recovered heat is used to drive absorption chillers that provide cooling through a DES chilled-water loop, or a portion of the recovered heat is used for a third, separate district energy source (e.g., if the building connects to both a steam loop and a hot-water loop), calculate the electricity generation assigned to each building using the formula for heat recovery-driven chillers.

$$\text{CHP_ELEC}_{\text{BLDG}} (\text{heat recovery-driven chillers}) = (X_{\text{HEAT}} \times \text{BLDG}_{\text{HEAT}}) + (Y_{\text{CHW}} \times \text{BLDG}_{\text{CHW}}) + (Z_{\text{SOURCE}} \times \text{BLDG}_{\text{SOURCE}}) \times \text{CHP_ELEC}_{\text{TOTAL}}$$

where

- $\text{CHP_ELEC}_{\text{BLDG}}$ = CHP electricity generation allocated to building
- X_{HEAT} = fraction of CHP plant's total production of waste heat applied to the DES directly
- $\text{BLDG}_{\text{HEAT}}$ = fraction of total district heat provided to building
- Y_{CHW} = fraction of CHP plant's total production of waste heat applied to producing chilled water in DES
- BLDG_{CHW} = fraction of total district chilled water provided to building
- Z_{SOURCE} = fraction of CHP plant's total production of waste heat applied to third district energy source
- $\text{BLDG}_{\text{SOURCE}}$ = fraction of third district energy source provided to building
- $\text{CHP_ELEC}_{\text{TOTAL}}$ = total CHP electricity generated at DES plant

When modeling CHP fuel input, allocate the CHP input fuel to the project building based on a proration and assignment of the total input fuel according to the results of the CHP electricity allocation described above for CHP electricity output. Use the energy cost and greenhouse gas emissions factors associated with the fuels input to the CHP. For the proposed case (all projects), calculate the CHP input fuel allocated to the building as follows:

$$\text{Proposed BLDG}_{\text{FUEL}} = \left(\frac{\text{CHP_ELEC}_{\text{BLDG}}}{\text{CHP_ELEC}_{\text{TOTAL}}} \right) \times \text{CHP}_{\text{FUEL}}$$

where

- $\text{Proposed BLDG}_{\text{FUEL}}$ = proposed case CHP input fuel allocated to building
- $\text{CHP_ELEC}_{\text{BLDG}}$ = CHP electricity generation allocated to building (from previous calculations)
- $\text{CHP_ELEC}_{\text{TOTAL}}$ = total CHP electricity generated at DES plant
- CHP_{FUEL} = total CHP fuel input for electricity generation at DES plant

For the baseline (scenario B in CHP electricity output only): calculate the CHP input fuel allocated to the building as follows:

$$\text{Baseline BLDG}_{\text{FUEL}} = \left(\frac{\text{PROCESS_ELEC}_{\text{BLDG}}}{\text{CHP_ELEC}_{\text{TOTAL}}} \right) \times \text{CHP}_{\text{FUEL}}$$

with

$$\text{PROCESS_ELEC}_{\text{BLDG}} = \text{CHP_ELEC}_{\text{BLDG}} - \text{PROPOSED_ELEC}_{\text{BLDG}}$$

where

- Baseline BLDG_{FUEL} = baseline case CHP input fuel charged to building
- PROCESS_ELEC_{BLDG} = amount of allocated CHP electricity in excess of building's modeled annual electricity consumption (treated as process energy in model)
- CHP_ELEC_{TOTAL} = total CHP electricity generated at DES plant
- CHP_{FUEL} = total CHP fuel input for electricity generation at DES plant
- CHP_ELEC_{BLDG} = CHP electricity generation allocated to building (from previous calculations)
- PROPOSED_ELEC_{BLDG} = modeled electricity consumption for building from proposed case

The model must include CHP generator efficiencies, based on either ongoing operations (existing CHP) or design specifications (new CHP).

Path 3 Large-scale District Energy System

Path 3 provides a streamlined method for documenting improved greenhouse gas (GHG) emissions performance associated with large scale district energy systems.

Complete the baseline and proposed energy modeling consistent with the guidance for Path 1A. ASHRAE 90.1-2016, Appendix G.

Follow the additional instructions in the *Large-Scale DES Calculator* (uploaded in the Credit Resources section of the credit library) to demonstrate GHG emissions improvement associated with the district energy plant.

Optional: Projects may generate two sets of baseline and proposed models to separately document the greenhouse gas emissions metric using Path 1A. ASHRAE 90.1-2016 Appendix G with the large-scale district energy calculator, and the cost metric using Path 1B. ASHRAE 90.1-2016 Appendix G with ASHRAE 90.1-2022 Addendum a.

Special Situations for DES Energy Models

Service water heating

If service water is heated in full or in part by DES-supplied heat: For projects applying Path 1A or Path 3, model the energy source as purchased energy.

Projects applying Path 1B shall model the DES supplied service water heating in the Proposed building per 90.1-2022 addendum a replacing all ASHRAE 90.1-2022 Addendum a references to Section 7 mandatory and prescriptive criteria for the *proposed building design* with ASHRAE 90.1-2016 Section 7.

The Baseline shall be modeled per 90.1-2016 Appendix G requirements.

For projects applying Path 2, model the baseline service water heating matching ASHRAE 90.1-2016 Appendix G modeling guidance for a stand-alone on-site service water heating system, and use the Path 2 guidance to model the average efficiency for the proposed design.

Heating converted to cooling

Sometimes the district or campus system heating energy supply is converted to chilled water using absorption chillers or other similar technologies to serve cooling loads. In this circumstance, the equipment that converts heating to cooling may reside within the DES itself, (i.e., DES provides cooling to the building) or within the connected buildings (i.e., DES provides heating to the building; building converts heating to cooling). When the equipment that converts DES-supplied heat into cooling is part of the LEED project's scope of work, the project must apply either Path 1B, Path 2, or Path 3.

Other DES systems

DES also often incorporate special features, such as thermal storage, ground or surface water cooling, and waste heat recovery. These features should be incorporated into the proposed virtual plant to the greatest extent practical using the general principles presented in this guidance.

Combined Heat and Power (CHP) or other Non-Renewable Electricity Generation Systems

For projects with combined heat and power or other non-renewable electricity generation systems, amend ASHRAE 90.1-2016 G2.4.2 Annual Energy Costs as follows:

Where the proposed design includes on-site electricity generation systems other than on-site renewable energy systems, adjust the baseline and proposed model using one of the following methods:

1. No credit for on-site electricity generation:
 - Model on-site electricity generation systems including all fuel inputs and associated site-recovered energy identically in the baseline design and the proposed design, OR
 - Model purchased electricity instead of the on-site electricity generation. Model any site-recovered energy from the on-site electricity generation system identically in the baseline and proposed design (either crediting it towards the thermal loads for both the baseline and proposed design or ignoring the site-recovered energy contribution in both the baseline and proposed design).
2. Credit for on-site electricity generation:

Model the baseline design using purchased electricity for all regulated energy sources except HVAC heating and/or service water heating modeled in accordance with Appendix G criteria. Model the proposed design to include the proposed on-site generation system including site-recovered energy. For the cost metric, natural gas or fuel rates for both the baseline and proposed design must be modeled using the current published rates for natural gas associated with the baseline design fuel usage excluding monthly meter charges and shall not be discounted for high fuel usage associated with on-site generation equipment.

International Tips

Option 1. Energy Performance Compliance

Canada: Use the Provincial emissions factors (where available) reported in the National Inventory Report, submitted by Canada to the United Nations Framework Convention on Climate Change, to calculate GHG emissions by energy source; these emissions factors are readily found in the ENERGY

STAR Portfolio Manager Greenhouse Gas Emissions Technical Reference
(<https://portfoliomanager.energystar.gov/pdf/reference/Emissions.pdf>).

Europe: The [Europe ACP for ASHRAE 90.1 Mandatory Provisions Table](#) provides further guidance for project teams in Europe wishing to use European standards in lieu of certain ASHRAE 90.1-2016 mandatory provisions in LEED v4.1. The guidance covers ASHRAE 90.1-2016 Mandatory Provision Sections 5.4, 6.4, 7.4, 8.4, 9.4 and 10. Column 1 of the table references the specific subsection used in ASHRAE 90.1-2016. Column 2 displays the requirement as written in ASHRAE 90.1-2016. Column 3 outlines the compliance pathway available for European projects. Column 4 includes, in some cases, further information about the proposal, differences between the proposal and the ASHRAE requirement, or a reference to further documentation.

Additionally, for projects using the Performance Option for compliance with EA prerequisite Minimum Energy Performance and EA credit Optimize Energy Performance, the documentation must also use the calculated U-factor for fenestration products including windows and skylights based on either the LBNL Windows 6 program, or a simulation software program that approximates the NFRC rating methodologies. Alternatively, a narrative shall be provided supporting the claim that the fenestration U-factor used in the model is similar to the values that would be achieved using the NFRC rating. The CE-marked fenestration does not account for thermal bridging and seasonal performance in the same way as the NFRC rating, and when accounted for in the energy model, has been observed to lead to savings that exceed those claimed for the same fenestration rated under the NFRC ratings.

Referenced Standards

- ▶ ASHRAE 90.1-2016
- ▶ ASHRAE 50% Advanced Energy Design Guides
- ▶ ASHRAE 209-2018
- ▶ [ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual, PNNL 2017](#)
- ▶ [Developing Performance Cost Index Targets for ASHRAE Standard 90.1 Appendix G - Performance Rating Method](#)

Required Documentation

Documentation	Energy Performance Option		
	90.1-2016 Prescriptive Compliance	90.1-2016 ECB Compliance (prerequisite only)	90.1 Appendix G Compliance
Minimum Energy Performance Calculator (90.1-2016) with Appendix G energy modeling inputs			X
Input-output reports from modeling software		X	X
Exceptional calculations (if applicable)			X
Energy consumption and demand for each building end use and fuel type			X
Description of energy utility rates for each energy source		X	X (if using cost metric)
Greenhouse gas emissions calculations, including emissions factors used			X (if using GHG metric)

ASHRAE 90.1-2016 Compliance Forms or COMcheck report confirming a system-by-system compliance with all mandatory and prescriptive provisions, and supporting construction drawings	X		
Documentation demonstrating compliance with ASHRAE 90.1-2016 Mandatory Measures and ECB		X	
Data center calculator (if applicable)			X
On-site renewable energy plans indicating location of renewable energy system, and relevant design details (e.g. PV module capacity, quantity, inverter capacity, tilt, orientation, etc. for a photovoltaic array), and confirming that the renewable energy is part of the project scope of work (or campus scope of work for a campus development)			X

Connection to Ongoing Building Performance

- LEED O+M EA credit Energy Performance: Designing building systems to achieve a minimum level of energy efficiency provides the foundation for effective energy management, reduced greenhouse gas emissions from building energy use, and reduced operating costs throughout the building life cycle.

EA Prerequisite: Building-Level Energy Metering

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To support energy management and identify opportunities for additional energy savings by tracking building-level energy use.

REQUIREMENTS

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Install new or use existing building-level energy meters, or submeters that can be aggregated to provide building-level data representing total building energy consumption (electricity, natural gas, chilled water, steam, fuel oil, propane, biomass, etc.). Utility-owned meters capable of aggregating building-level resource use are acceptable.

Commit to sharing with USGBC the resulting energy consumption data and electrical demand data (if metered) for a five-year period beginning on the date the project accepts LEED certification. At a minimum, energy consumption must be tracked at one-month intervals.

This commitment must carry forward for five years or until the building changes ownership or lessee.

CS

Install new or use existing base building-level energy meters, or submeters that can be aggregated to provide base building-level data representing total building energy consumption (electricity, natural gas, chilled water, steam, fuel oil, propane, etc.). Utility-owned meters capable of aggregating base building-level resource use are acceptable.

Commit to sharing with USGBC the resulting energy consumption data and electrical demand data (if metered) for a five-year period beginning on the date the project accepts LEED certification or typical occupancy, whichever comes first. At a minimum, energy consumption must be tracked at one-month intervals.

This commitment must carry forward for five years or until the building changes ownership or lessee.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Installing whole building energy meters is a simple yet critical strategy for understanding total energy use throughout the building life cycle. Tracking energy consumption on a regular basis supports effective energy management and provides data to help verify that building systems are operating as designed. Projects can submit data via the Arc platform to comply with the prerequisite requirement to share whole-project energy usage data with USGBC and get started on the path to recertification.

EA Prerequisite: Fundamental Refrigerant Management

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce ozone depletion and global warming potential and support early compliance with the Kigali Amendment to the Montreal Protocol, while minimizing direct contributions to climate change.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Do not use chlorofluorocarbon (CFC) or hydro chlorofluorocarbon (HCFC) -based refrigerants in new heating, ventilating, air-conditioning, and refrigeration (HVAC&R) systems. When reusing existing HVAC&R equipment, complete a comprehensive CFC and/or HCFC phase-out conversion before project completion. Phase-out plans extending beyond the project completion date will be considered on their merits.

Existing small HVAC&R units (defined as containing less than 0.5 pound [225 grams] of refrigerant) and other equipment, such as standard refrigerators, small water coolers, and any other equipment that contains less than 0.5 pound (225 grams) of refrigerant, are exempt.

GUIDANCE

Refer to the LEED v4 reference guide, with the following edits and additions:

Behind the Intent

Hydrochlorofluorocarbons (HCFCs) have shorter atmospheric lives than CFCs, but they still contribute to ozone depletion. Similar to CFCs, HCFCs have phase-out dates scheduled under the Montreal Protocol.

Both HCFCs and CFCs must be addressed to meet this prerequisite. Wherever CFCs are mentioned in the LEED v4 reference guide, the guidance also applies to HCFCs.

Further Explanation

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Enhanced Refrigerant Management: Despite successful global adoption of the Montreal Protocol, large volumes of CFCs and HCFCs remain in circulation and contribute to stratospheric ozone depletion. Completing a comprehensive CFC phase-out conversion for HVAC&R equipment reused in the project helps ensure that refrigerants are correctly disposed of and do not escape into the atmosphere.

EA Credit: Enhanced Commissioning

This prerequisite applies to

- ▶ BD+C: New Construction (2-6 points)
- ▶ BD+C: Core & Shell (2-6 points)
- ▶ BD+C: Schools (2-6 points)
- ▶ BD+C: Retail (2-6 points)
- ▶ BD+C: Data Centers (2-6 points)
- ▶ BD+C: Warehouses & Distribution Centers (2-6 points)
- ▶ BD+C: Hospitality (2-6 points)
- ▶ BD+C: Healthcare (2-6 points)

INTENT

To further support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Implement, or have in place a contract to implement, the following commissioning process activities in addition to those required under EA Prerequisite Fundamental Commissioning and Verification.

Commissioning Authority Qualifications:

- ▶ The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- ▶ The CxA may be a qualified employee of the owner, an independent consultant, an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.

Option 1. Enhanced Systems Commissioning (3-4 points)

Path 1: Enhanced Commissioning (3 points)

Complete the following commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies in accordance with ASHRAE Guideline 0-2013 and ASHRAE Guideline 1.1-2007 for HVAC&R systems, as they relate to energy, water, indoor environmental quality, and durability.

The commissioning authority must do the following:

- ▶ Review contractor submittals.
- ▶ Verify inclusion of systems manual requirements in construction documents.
- ▶ Verify inclusion of operator and occupant training requirements in construction documents.
- ▶ Verify systems manual updates and delivery.
- ▶ Verify operator and occupant training delivery and effectiveness.
- ▶ Verify seasonal testing.
- ▶ Review building operations 10 months after substantial completion.
- ▶ Develop an on-going commissioning plan.

Include all enhanced commissioning tasks in the OPR and BOD.

OR

Path 2: Enhanced and Monitoring-Based Commissioning (4 points)

Achieve Path 1.

AND

Develop monitoring-based procedures and identify points to be measured and evaluated to assess performance of energy- and water-consuming systems.

Include the procedures and measurement points in the commissioning plan. Address the following:

- ▶ roles and responsibilities;
- ▶ measurement requirements (meters, points, metering systems, data access);
- ▶ the points to be tracked, with frequency and duration for trend monitoring;
- ▶ the limits of acceptable values for tracked points and metered values (where appropriate, predictive algorithms may be used to compare ideal values with actual values);
- ▶ the elements used to evaluate performance, including conflict between systems, out-of-sequence operation of systems components, and energy and water usage profiles;
- ▶ an action plan for identifying and correcting operational errors and deficiencies;
- ▶ training to prevent errors;
- ▶ planning for repairs needed to maintain performance; and
- ▶ the frequency of analyses in the first year of occupancy (at least quarterly).

Update the systems manual with any modifications or new settings, and give the reason for any modifications from the original design.

AND/OR

Option 2. Building Enclosure Commissioning (2 points)

Fulfill the requirements in EA Prerequisite Fundamental Commissioning and Verification as they apply to the building's enclosure in addition to mechanical and electrical systems and assemblies.

Complete the following commissioning (Cx) process activities for the building's thermal envelope in accordance with ASHRAE Guideline 0-2013 and ASTM E2947-16: Standard Guide for Building Enclosure Commissioning, as they relate to energy, air and water tightness, indoor environmental quality, and durability.

The qualified independent member of the design or construction team responsible for building enclosure commissioning must complete the following:

- ▶ Review contractor submittals.
- ▶ Verify inclusion of systems manual requirements in construction documents for enclosure systems.
- ▶ For specialty enclosure systems with controls and automation:
 - Verify inclusion of operator and occupant training requirements in construction documents.
 - Verify systems manual updates and delivery.
 - Verify operator and occupant training delivery and effectiveness.
 - Verify seasonal testing.
 - Review building operations 10 months after substantial completion.
- ▶ Develop an on-going enclosure commissioning plan for maintenance, renewal, and revitalization cycles.

Data Centers only

Projects that select Option 1 must complete the following commissioning process.

For small projects with peak cooling loads less than 2,000,000 Btu/h (600 kW), or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA must perform the following activities:

- ▶ conduct at least one commissioning verification review of the owner's project requirements, basis of design, and design documents before mid-construction documents development;
- ▶ back-check the review comments in all subsequent design submissions; and
- ▶ conduct an additional full verification review at 95% completion of the design documents and basis of design.

For projects with peak cooling loads 2,000,000 Btu/h (600 kW) or more, or a total computer room peak cooling load 600,000 Btu/h (175 kW) or more, the CxA must conduct at least three verification reviews of the basis of design:

- ▶ one verification review of design documents before the start of design development;
- ▶ one verification review of design documents before midconstruction documents; and
- ▶ one final verification review of 100% complete design documents, verifying achievement of the owner's project requirements and adjudication of previous review comments.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications.

Behind the Intent

Beta Update

Updated referenced standards. Minor text revisions to Option 2. Building Enclosure Commissioning intend to clarify implementation for project teams.

Commissioning Authority criteria are expanded to include an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.

Further Explanation

Building Enclosure Commissioning Basics

Refer to the LEED v4 reference guide, with the following modification:

- ▶ See ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning and ASTM E2813-12: Standard Practice For Building Enclosure Commissioning for additional guidance regarding envelope systems testing equipment and procedures.

Referenced Standards:

- ▶ ASHRAE Guideline 0-2013, The Commissioning Process
- ▶ ASHRAE Guideline 1.1-2007, HVAC&R Technical Requirements for the Commissioning Process
- ▶ ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning
- ▶ ASTM E2813-12: Standard Practice for Building Enclosure Commissioning

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Enhanced building commissioning expands the fundamental commissioning process to provide further oversight and verification of mechanical systems, which ensures ongoing building quality control and operations and may help improve the project's energy performance score. Monitoring-based commissioning is a powerful strategy for tracking building performance in real time. Building enclosure commissioning provides quality assurance for enclosure systems design and installation and provides the foundation for energy efficiency throughout the building life cycle.

EA Credit: Optimize Energy Performance

This prerequisite applies to

- ▶ BD+C: New Construction (1-18 points)
- ▶ BD+C: Core & Shell (1-18 points)
- ▶ BD+C: Schools (1-16 points)
- ▶ BD+C: Retail (1-18 points)
- ▶ BD+C: Data Centers (1-18 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-18 points)
- ▶ BD+C: Hospitality (1-18 points)
- ▶ BD+C: Healthcare (1-20 points)

INTENT

To achieve increasing levels of energy performance beyond the prerequisite standard to reduce environmental and economic harms associated with excessive energy use and greenhouse gas emissions that disproportionately impact frontline communities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Analyze efficiency measures during the design process and account for the results in design decision making. Use energy simulation of efficiency opportunities, past energy simulation analyses for similar buildings, or published data (e.g., Advanced Energy Design Guides) from analyses for similar buildings.

Analyze efficiency measures, focusing on load reduction and HVAC-related strategies (passive measures are acceptable) appropriate for the facility. Project potential energy savings and holistic project cost implications related to all affected systems.

Choose one of the options below.

Option 1. Energy Performance Compliance (1-18 points except Schools and Healthcare, 1-16 points Schools, 1-20 points Healthcare)

Demonstrate a Performance Cost Index (PCI)¹ below the Performance Cost Index Target (PCI_t) calculated in accordance with Section 4.2.1.1 of ANSI/ASHRAE/IESNA Standard 90.1-2016, Appendix G, Table 4.2.1.1. For mixed use buildings, the required PCI shall be calculated by using an area weighted average of the building types.

Calculate the PCI, PCI_t, and percentage improvement using metrics of cost and greenhouse gas (GHG) emissions.

Total LEED points for EAc Optimize Energy Performance are determined by summing the points documented in Table 1 (energy cost metric) plus the points documented in Table 2 (GHG emissions metric), where points for each Table are determined based on the project percent improvement PCI below the PCI_t documented for each metric.

Table 1. Points for percentage improvement in energy performance – Percent Cost PCI below PCI_t (1-9 points NC and CS, 1-8 points Schools, 1-10 points Healthcare)

New Construction	Healthcare, Major Renovation, CS	Points BD+C (except Schools, Healthcare)*	Points Healthcare	Points Schools
5%	2%	1	1	1
10%	5%	2	2	2
15%	10%	3	3	3
20%	15%	4	4	4
25%	20%	5	5	5
30%	25%	6	6	6
35%	30%	7	7	7
40%	35%	8	8	
45%	40%	9	9	8
50%	45%	EP	10	EP
	50%		EP	

Table 2. Points for percentage improvement in energy performance – Percent Greenhouse Gas Emissions PCI below PCI_t (1-9 points NC, 1-8 points Schools, 1-10 points Healthcare)

New Construction	Healthcare, Major Renovation, CS,	Points BD+C (except Schools, Healthcare)*	Points Healthcare	Points Schools
5%	2%	1	1	1
10%	5%	2	2	2
16%	10%	3	3	3
24%	16%	4	4	4
32%	24%	5	5	5
40%	32%	6	6	6
50%	40%	7	7	7
65%	50%	8	8	
80%	65%	9	9	8
100%	80%	EP	10	EP
			EP	

Table 1 and Table 2 Notes:

1. BD+C projects except Data Centers with Proposed Building Unregulated Energy exceeding 50% of the total Proposed Building Performance for the referenced metric (cost or GHG emissions), and BD+C: Data Centers projects with at least 40% gross colocation data center area may use the “Healthcare, Major Renovation, CS” column in lieu of the “New Construction” column.
2. Renewable Energy:
 - On-site renewable energy: For Table 1 and Table 2, on-site or on-campus renewable energy that meets ASHRAE Standard 90.1-2016 Section G 2.4.1 requirements for on-site renewable may be subtracted from the Proposed Building Performance in accordance with ASHRAE Standard 90.1-2016 Section G 2.4.1.
 - For the Table 2 greenhouse gas emissions metric, Tier 2 New off-site renewables documented in EA Credit Renewable Energy may be subtracted from proposed greenhouse gas emissions for electricity prior to calculating proposed building performance. Projects in locations where hourly electric greenhouse gas emissions factors are available must document performance for the baseline, proposed, and renewable generation using hourly electric emissions factors.

DATA CENTERS ONLY

In addition to the requirements above, analyze efficiency measures focused on IT load reduction and HVAC-related strategies. Collocated data centers may use the CS percentage improvement thresholds in lieu of NC thresholds to earn points in Table 1 and Table 2.

OR

Option 2. Prescriptive Compliance: ASHRAE Advanced Energy Design Guide (1-6 points, 1-4 points CS)

To be eligible for Option 2, projects must meet the Scope requirements of the applicable AEDGs (or combination of AEDGs for mixed use), and projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance.

Implement and document compliance with the applicable recommendations and standards in Chapter 4, Design Strategies and Recommendations by Climate Zone, for the appropriate ASHRAE 50% Advanced Energy Design Guide and climate zone.

ASHRAE 50% Advanced Energy Design Guide for Small to Medium Office Buildings

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- ▶ *Interior and exterior lighting*, including daylighting and interior finishes (1 point).
- ▶ *Plug loads*, including equipment and controls (2 points, 0 points CS)
- ▶ *HVAC Systems and Controls* (2 points, 1 point CS).

ASHRAE 50% Advanced Energy Design Guide for Medium to Large Box Retail Buildings

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, vestibules, and fenestration – all orientations (1 point, 2 points CS)
- ▶ *Interior and exterior lighting*, excluding lighting power density for sales floor (1 point).
- ▶ *Additional interior lighting* for sales floor (1 point, 0 points CS)
- ▶ *Plug loads*, including equipment choices and controls (1 point, 0 points CS)
- ▶ *HVAC efficiency and control requirements* (2 points, 1 point CS).

ASHRAE 50% Advanced Energy Design Guide for K-12 School Buildings

(Not applicable for CS)

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, and vertical fenestration (1 point)
- ▶ *Interior and exterior lighting*, including daylighting and interior finishes (1 point)
- ▶ *Plug loads*, including equipment choices, controls, and kitchen equipment (2 points)
- ▶ *HVAC efficiency and control requirements* (2 points)

ASHRAE 50% Advanced Energy Design Guide for Large Hospitals

- ▶ *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- ▶ *Interior and exterior lighting,* including daylighting (form or nonform driven) and interior finishes (1 point).
- ▶ *Plug loads,* including equipment choices, controls, and kitchen equipment (1 point, 0 points CS)
- ▶ *HVAC and Service Water Heating Systems and Equipment (2 points) (1 point CS.)*

ASHRAE 50% Advanced Energy Design Guide for Grocery Stores

- ▶ *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point) (2 points – CS)
- ▶ *Interior and exterior lighting, including sales floor (1 points)*
- ▶ *Refrigeration, Plug, and Process loads,* including equipment choices and controls (2 points) (0 points CS)
- ▶ *HVAC efficiency and control requirements (1 point) (1 point CS.)*

Option 3. Systems Optimization (1-6 points)

To be eligible for Option 3, projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance, and must not have more than 2,000 square feet of data center space, laboratory space, or manufacturing space.

Demonstrate an improvement beyond ASHRAE/ASHRAE/IESNA Standard 90.1-2016, with errata, for the following systems as applicable to the project scope: Interior and Exterior Lighting; Daylight controls; Building envelope; HVAC and service water heating equipment efficiency; and Equipment and appliances.

Use any combination of the strategies in any or all of the categories below, for a maximum of up to 6 points.

- ▶ *Interior and Exterior Lighting:*
 - 15% lighting power reduction (1 point)
 - 30% lighting power reduction (1 point)
 - 45% lighting power reduction (1 point)
- ▶ *Daylight controls:*

Install daylight-responsive controls for a given percentage of connected lighting load (lighting in non-regularly occupied space with occupant sensor controls may be excluded from connected lighting load).

 - 40% (1 point)
- ▶ *Building envelope:*
 - Climate Zones 1 – 2: Achieve 2 of the 3 strategies below for 1 point.
 - Thermal Mass Enclosure: More than 70% of opaque above-grade wall area meets ASHRAE 90.1-2016 definition for “mass wall”; and more than 70% of floor area meets ASHRAE 90.1-2016 definition for “mass floor”
 - 25% Envelope UA reduction
 - 50% SHGC reduction (including window shade factors)
 - Climate Zones 3 – 8: Achieve 2 of the 3 strategies below for 1 point.
 - 25% Envelope UA reduction
 - 50% Envelope UA reduction
 - 25% reduction in air infiltration measured during commissioning
- ▶ *HVAC and Service Water Heating Equipment Efficiency:*
 - Reduction in total fan power allowance of:
 - 20% (1 point)
 - Improvement in efficiency for at least 75% of the combined cooling, heating, and service water heating capacity
 - 20% (1 point)

- *Electric resistance heating except heat pump auxiliary heat must be included in total capacity.*
- ▶ **Equipment and Appliances:**
Install a percentage (by rated power) of eligible equipment and appliances meeting the following requirements:
 - *ENERGY STAR* equipment including appliances, office equipment, electronics, and commercial food service equipment (lighting and building envelope products are excluded from this credit). (Electronic Product Environmental Assessment Tool (EPEAT) equipment may be used in lieu of Energy Star equipment where applicable).
 - Prescriptive commercial kitchen and refrigeration equipment requirements listed in Appendix 3, Table 1.

The project scope of work must include at least 0.25 Watts per square foot of eligible equipment to apply this strategy.

Percent of Eligible Equipment Installed by Rated Power:

- ▶ 75% (1 point)

OR

Option 4. Data Centers only- System Optimization (1-3 points)

Calculate an Overall Systems Design Value as the sum of the maximum *design Mechanical Load Component (MLC)* and maximum *design Electrical Load Component (ELC)* in accordance with ASHRAE 90.4-2016 Section 6.2, Section 8.2, and Section 11. Document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 10% (1 point)
- 20% (2 points)
- 30% (3 points)

**If the electrical system design is incomplete, the design values shall be assumed to match the values in Table 8.2.1.1 and 8.2.1.2.*

Collocated data centers: document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 6% (1 point)
- 12% (2 points)
- 18% (3 points)

GUIDANCE

Refer to the consolidated guidance within EA Prerequisite Minimum Energy Performance, since the requirements and LEED documentation are closely linked for EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance:

Behind the Intent

See EA Prerequisite Minimum Energy Performance, *Behind the Intent*.

Beta Update

Updated referenced standards and a new greenhouse gas emissions metric ensure that LEED continues to be a global leadership standard for energy performance and encourages owners to directly consider and address building carbon emissions. Revised prescriptive options to expand applicability to a

broader set of projects. Require design phase analysis for all Optimize Energy Performance options, including prescriptive and performance options.

Step-by-Step Guidance

See EA Prerequisite Minimum Energy Performance.

Assure that the additional Optimize Energy Performance requirement credit requirements are met:

- Analyze efficiency measures during the design process, focusing on load reduction and HVAC-related strategies (including passive measures) appropriate for the facility, project potential energy savings and holistic project cost implications, and account for the results in design decision making.

These requirements apply regardless of whether the project is using the performance path (Option 1) or prescriptive path (Options 2, 3 or 4) to document credit compliance.

The best approach for analyzing efficiency measures is a preliminary energy model, which evaluates heating and cooling load reduction strategies, passive HVAC strategies, and HVAC efficiency and control strategies (see EA prerequisite Minimum Energy Performance, *Further Explanation, Developing a Preliminary Energy Model* and *Further Explanation, Modeling HVAC Systems*). ASHRAE Standard 209 provides a standardized methodology which may be used for developing preliminary energy models that are used to inform the design process (See ASHRAE 209 Sections 6.3 - Load Reduction Modeling and 6.4 - HVAC System Selection Modeling).

Projects may also opt to use the AEDGs where applicable, and/or use modeling analyses previously performed for similar projects to meet the design phase analysis requirements.

Further Explanation

Calculations

See EA Prerequisite Minimum Energy Performance, *Further Explanation, Calculations* for calculation of the Performance Cost Index (PCI) and Performance Cost Index Target (PCI_t) using units of cost and GHG emissions.

Applying Renewable Energy Savings

See EA Prerequisite Minimum Energy Performance, *Further Explanation, Applying Renewable Savings* Only projects pursuing Option 1 of this credit may count savings from renewable energy systems.

Rating System Variations

Data Centers

See EA prerequisite Minimum Energy Performance, *Further Explanation, Rating System Variations*.

Co-located data centers may use the percentage improvement thresholds for Healthcare, Major Renovation, and CS in Table 1. Points for percentage improvement in energy performance - % Cost PCI below PCI_t and in Table 2. Points for percentage improvement in energy performance - % Greenhouse gas emissions PCI below PCI_t to determine points achieved under this credit.

Referenced Standards

- ▶ ASHRAE 90.1-2016
- ▶ ASHRAE 90.4-2016
- ▶ ASHRAE 50% Advanced Energy Design Guides
- ▶ ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual, PNNL 2017
- ▶ ASHRAE 209-2018
- ▶ Developing Performance Cost Index Targets for ASHRAE Standard 90.1 Appendix G - Performance Rating Method

Exemplary Performance

Option 1.

- ▶ Achieve 50% savings in the cost metric (45% for CS, Major Renovation, and Multifamily Residential), or
- ▶ Achieve 100% savings in the greenhouse gas emissions metric, (80% for CS, Major Renovation, and Multifamily Residential), or
- ▶ Use actual utility rates including all applicable seasonal, TOU, and demand components when calculating cost savings AND hourly greenhouse gas emissions factors when calculating greenhouse gas emissions savings.

Required Documentation

In addition to the documentation required for EA prerequisite Minimum Energy Performance, the following documentation is required for credit compliance

Documentation	Option 1	Option 2	Option 3	Option 4
Confirmation that energy efficiency measures were analyzed during the design phase and that the results were considered in design decision-making. •	X	X	X	X
Target Finder results and summary	X (for applicable building types)			
Greenhouse gas emissions calculations that include credit for new off-site renewable energy (as applicable)	X (where applicable)			
AEDG compliance tables		X		
Calculations demonstrating achievement of systems optimization requirements (e.g. ASHRAE 90.1 lighting power density calculations, lighting fixture calculations demonstration portion of lighting power with daylighting control, etc.)			X	
ASHRAE 90.4 Calculations (consistent with documentation requirements listed in ASHRAE 90.4)				X

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Designing for higher levels of energy efficiency helps decrease building operating costs and reduces the environmental and economic harms associated with excessive energy use throughout the building life cycle; this may help improve the building's energy performance score.

EA Credit: Advanced Energy Metering

This prerequisite applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To support energy management and identify opportunities for additional energy savings by tracking building-level and system-level energy use.

REQUIREMENTS

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Install *advanced energy metering* for the following:

- ▶ all whole-building energy sources used by the building; and
- ▶ any individual energy end uses that represent 10% or more of the total annual consumption of the building.

The advanced energy metering must have the following characteristics.

- ▶ Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.
- ▶ Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- ▶ The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- ▶ The system must be capable of storing all meter data for at least 36 months.
- ▶ The data must be remotely accessible.
- ▶ All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

CS

Install meters for future tenant spaces so that tenants will be capable of independently metering energy consumption (electricity, chilled water, etc.) for all systems dedicated to their space. Provide a sufficient number of meters to capture total tenant energy use with a minimum of one meter per energy source per floor.

Install *advanced energy metering* for all base building energy sources used by the building.

The advanced energy metering must have the following characteristics.

- ▶ Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.
- ▶ Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- ▶ The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- ▶ The system must be capable of storing all meter data for at least 36 months.
- ▶ The data must be remotely accessible.

- ▶ All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Submetering energy subsystems helps facility managers track changes in energy usage over time and provides the performance data necessary to identify opportunities for energy savings by end use. Submetering is an important component of a successful energy management program; metered data enables monitoring of consumption and costs as well as progress reporting throughout the building life cycle.

EA Credit: Grid Harmonization

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To increase participation in demand response technologies and programs that make energy generation and distribution systems more affordable and more efficient, increase grid reliability, and reduce greenhouse gas emissions.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Design building and equipment for participation in demand response programs through load shedding or shifting. On-site electricity generation does not meet the intent of this credit.

Case 1. Demand Response Program Available and Participation (2 points)

- ▶ Participate in an existing demand response (DR) program and complete the following activities. Design a system with the capability for real-time, fully-automated DR based on external initiation by a DR Program Provider. Semi-automated DR may be utilized in practice.
- ▶ Enroll in a minimum one-year DR participation amount contractual commitment with a qualified DR program provider, with the intention of multiyear renewal, for at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance. The on-peak demand may vary based on the utility climate and pricing structures.
- ▶ Develop a comprehensive plan for meeting the contractual commitment during a Demand Response event.
- ▶ Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- ▶ Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- ▶ Initiate at least one full test of the DR plan.

OR

Case 2. Demand Response Capable Building (1 point)

Have infrastructure in place to take advantage of future demand response programs or dynamic, real-time pricing programs and complete the following activities:

- ▶ Install interval recording meters and have equipment capable of accepting an external signal.
- ▶ Develop a comprehensive plan for shedding at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance.
- ▶ Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.

- ▶ Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- ▶ Contact local utility representatives to discuss participation in future DR programs.

AND/OR

Case 3. Load Flexibility and Management Strategies (1-2 points)

Analyze the building's annual load shape and peak load based as calculated for EA prerequisite Minimum Energy Performance. Review the regional grid load profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's (EPA) AVOIDed Emissions and geneRation Tool (AVERT) provides regional grid emissions data; local utilities may also provide this data.

Coordinate review of building load shape and peak load with review of the regional grid profile to identify the best value load management strategies that the building can provide.

Implement one or more of the load flexibility and management strategies described below. All projects must install interval recording meters and have equipment capable of accepting an external signal.

Load Flexibility and Management Strategies:

- ▶ Peak Load Optimization: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand referenced to the ASHRAE 90.1-2016 compliant case (1 point)
- ▶ On-site thermal and/or electricity storage: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)

Include installed technology in the scope of work for the commissioning authority. Include load flexibility and management strategies and installed technologies in the building systems manual, or include in the current facilities requirements and operations maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.

Contact local utility representatives to discuss participation in future DR programs and to inform utility of building load flexibility and management strategies.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

As the number of distributed energy resources, or grid-connected power generation from individual buildings, increases, the utility must integrate these resources while continuing to manage overall grid capacity and generation. Too much distributed generation that is not integrated into the energy system can impair grid operation. Buildings have the opportunity to support effective grid management by designing on-site electricity systems that are integrated components of the energy system, serving as resources for the grid rather than disruptors.

In addition to participation in a utility's demand response program, projects can install storage technologies and implement operational strategies that support effective grid management and increase grid reliability. Eligible technologies may include battery storage, flywheel energy storage, or thermal energy storage; load management strategies may include load shifting or flexible operating scenarios.

Beta Update

Updates intend to address projects where no demand response program is available from the local utility and recognize new distributed energy resources technologies and grid harmonization strategies.

Step-by-Step Guidance

Refer to Steps 1 – 5 in the LEED v4 reference guide, with the following addition:

Step 6. Include demand response in commissioning and in building systems manual
Coordinate with the CxA to include a review of the DR plan in the commissioning of the building's system test procedures, to verify the ability to handle an externally initiated demand response event. The Cx plan must include at least one performance test of the full DR plan to verify that all equipment responds as planned and that all responsible parties understand their roles.

Include demand response program and any installed technologies in the building systems manual.

Case 1. Demand Response Program Available

Refer to the LEED v4 reference guide EAc Demand Response, with the following addition:

Step 2. Initiate at least one full test of the DR plan.
Execute at least one performance test of the full DR plan as defined in the Cx plan. All relevant DR operations team members must participate, and address and mitigate any issues identified as a result of the performance test.

Case 2. Demand Response Capable Building

Refer to the LEED v4 reference guide, Case 2. Demand Response Program Not Available, with the following change:

Under the LEED v4.1 update, project eligibility for Case 2 is no longer restricted to applications where a demand response program is not available. However, the project team must still contact the utility service provider to discuss potential participation in a current or future DR program.

Case 3. Load Flexibility and Management Strategies

Step 1. Analyze building load shape

Analyze building load shape as calculated for EA prerequisite Minimum Energy Performance; compare building load shape with the regional grid profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's AVOIDed Emissions and geneRation Tool (AVERT) provides hourly regional grid emissions data or contact the local utility to request this data.

Step 2. Identify highest-value strategies

Based on the analysis of building load shape, peak load and regional grid peak load or peak carbon emissions, identify storage technologies or load management strategies that the building can implement at the lowest cost while providing the greatest value to the grid.

Step 3. Implement strategies

Implement one or both of the strategies identified during step 2.

Eligible strategies include peak load optimization achieved by reducing on-peak load by at least 10% compared to peak electrical demand, or installation of on-site thermal energy storage for heating and cooling and/or electricity storage capable of reducing on-peak load by at least 10% as compared to peak electrical demand. On-site storage enables the building to store energy and use it during peak demand times, increasing annual energy savings and reducing strain on the grid.

Step 4. Include load flexibility and management system in commissioning and in building systems manual

Coordinate with the CxA to include a review of the load management system and supporting technologies in the commissioning of the building's system test procedures. The Cx plan must include at least one performance test of the full system to verify that all equipment operates as planned and that all responsible parties understand their roles.

Include the system and any installed technologies in the building systems manual.

Step 5. Contact local utility

Contact the local utility or service provider to express interest in participation in a future demand response program, and to inform the utility of building load flexibility and management strategies.

Further Explanation

On-Peak Electricity Demand

On-peak electricity demand can be determined as one of the following for the purposes of credit compliance:

1. Building peak electricity demand
2. Building demand load coinciding with regional grid peak demand
3. Building demand load coinciding with regional grid peak carbon emissions

Grid Harmonization

Energy efficient and grid-interactive buildings save money and resources while supporting broader grid-scale decarbonization. Savings to building owners accrue from a combination of demand charge reductions, lower annual energy use, utility incentives, and increased grid resilience.

There are a number of methods by which projects can comply with the requirement in Case 2 and Case 3 to install equipment capable of responding to an external signal. External signals can be sent from a variety of sources such as the electric utility, the electricity Independent System Operator (ISO), or a third-party automation organization. The signal can be used to indicate specific events such as a critical price peaks, grid reliability events, high grid demand, or high grid GHG emissions factors; or a continuous signal reporting price, reliability, GHG emissions, or other factors. This equipment may be part of a building automation system, stand-alone equipment, or equipment that is directly integrated with the building system(s) used to achieve the credit, e.g. a packaged rooftop HVAC system.

Peak Load Optimization

To demonstrate that the peak load optimization strategy reduces on-peak load by at least 10% as compared to peak electrical demand, the baseline should be referenced to the ASHRAE 90.1-2016 compliant case. The "Performance Demand Index Target" and "Performance Demand Index" shall be calculated using the same methodology as the PCIT and PCI calculated per ASHRAE 90.1-2016 Appendix G:

$$\text{Performance Demand Index Target} = [\text{Baseline Building Unregulated Electric Demand} + (\text{BPF} \times \text{Baseline Building Regulated Electric Demand})] / \text{Baseline Building Demand}$$

For example, if the ASHRAE 90.1-2016 Building Performance Factor is 0.58, the Baseline regulated demand is 75 kW, and the Baseline unregulated demand is 25 kW for an Office building in climate zone 4A, the Performance Demand Index Target would be calculated as:

$$\text{Performance Demand Index Target} = [25 + (0.58 \times 75)] / (25 + 75) = 0.69$$

The proposed demand would need to show a 10% improvement beyond this Performance Demand Target. Therefore, for this application:

$$\text{Proposed demand} \leq (0.69 \times 100) \times 90\% = 62.1 \text{ kW}$$

In cases where the proposed case incorporates electrification of space heating and/or service water heating compared to the baseline case, project teams have two options for demonstrating peak load optimization:

- ▶ If able to demonstrate 10% on-peak load reduction with fuel switching, the project team may complete the calculation without applying corrections to the baseline.
- ▶ If unable to demonstrate 10% on-peak load reduction with fuel switching, the project team may increase the baseline building demand by the baseline case fossil fuel heating and service water heating demand divided by 2.0.

On-site Thermal and/or Electricity Storage

The reference case for demonstrating peak demand reduction via the On-site thermal and/or electricity storage strategy is the Proposed design from the Optimize Energy Performance energy model without the thermal and/or electricity storage strategy in place.

Single Strategy, Multiple Points

In some cases, a single design or operational strategy may meet the criteria for more than one strategy. For example, a set of demand reduction strategies used to achieve Case 2. Demand Response Capable Building may also meet the criteria for Case 3. Load Flexibility and Management Strategies for Peak Load Optimization.

Project Type Variations

District Energy Systems

When district energy is modeled for credit in EA credit Optimize Energy Performance, the district energy system (DES) electric demand modeled there must be included in the total demand reported for this credit. When the DES system is included in the EA credit Optimize Energy Performance model, grid harmonization strategies applied to the DES system may be used to document achievement at the building level.

The interval recording meter and building automation system communications may be located in the DES – and not in the building – if strategies in the DES are used to demonstrate full achievement of the credit requirements at the building level.

Required Documentation

Documentation	Case 1	Case 2	Case 3
Proof of enrollment in DR program	X		
Evidence of ability to shed 10% of on-peak demand	X	X	X
Confirmation that system is capable of receiving and acting on external signal	X	X	X
Action plan for meeting reduction requirement during event	X	X	X
Inclusion of DR in CxA systems testing plan	X	X	X
Inclusion of DR and/or grid harmonization technologies in building systems manual	X	X	X
Documentation of one full test of the DR plan	X		
Narrative or report that includes: summary of building annual load shape and regional grid profile analysis; description of building load flexibility and/or management strategies implemented			X
Documentation of technologies serving the project, as applicable			X

Connection to Ongoing Building Performance

- Participation in a demand response program enables projects to support efficient energy generation and distribution systems, increase grid reliability, and reduce greenhouse gas emissions. For projects where no demand response program is available, implementing load flexibility and management strategies can help to achieve the same outcomes and support effective electrical grid management.

Exemplary Performance

Achieve Case 1 with the capability for at least a 20% peak demand savings, or achieve at least three points under Case 1 and Case 3, or achieve at least three points under Case 2 and Case 3.

EA Credit: Renewable Energy

This prerequisite applies to

- ▶ BD+C: New Construction (1-5 points)
- ▶ BD+C: Core & Shell (1-5 points)
- ▶ BD+C: Schools (1-5 points)
- ▶ BD+C: Retail (1-5 points)
- ▶ BD+C: Data Centers (1-5 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-5 points)
- ▶ BD+C: Hospitality (1-5 points)
- ▶ BD+C: Healthcare (1-5 points)

INTENT

To reduce the environmental and economic harms associated with fossil fuel energy and reduce greenhouse gas emissions by increasing the supply of renewable energy projects and foster a just transition to a green economy.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Use on-site renewable energy systems or procure renewable energy from offsite sources for all or a portion of the building's annual energy use.

Choose one or more strategies for renewable energy procurement from the categories below. Points achieved in each category may be added for a total of 5 points.

- ▶ **Tier 1: On-site renewable energy generation**
 - On-site renewable energy generation, environmental attributes (e.g. RECs) retained
- ▶ **Tier 2: New off-site renewable energy**
 - Off-site renewable electricity that is produced by a generation asset(s) built within the last five years or contracted to be operational within two years of building occupancy.
 - Green-e Energy certification or equivalent is required for one-time purchase and delivery of EACs of more than 100% of the project's annual electricity use.
- ▶ **Tier 3: Off-site renewable energy**
 - Off-site renewable electricity that is Green-e Energy certified or equivalent or captured bio-methane

Ownership of Environmental Attributes: All environmental attributes associated with renewable energy generation must be retired on behalf of the LEED project in order for the renewable energy procurement to contribute to credit achievement.

The default contract length for renewable energy procurement is 10 years. Contract lengths less than 10 years may be pro-rated.

For Tier 2, the age of the generation asset(s) is assessed at the beginning of the contract, and the generation asset(s) retain this attribute for the duration of the initial contract or lease term.

Off-site renewable energy must be generated by renewable electricity projects located in the same country or region where the LEED project is located. Methane capture in the form of biogas that is both captured and used on-site may qualify as a Tier 1 renewable energy resource.

Points are awarded according to Table 1, based on the percentage of total site energy use. Renewable electricity and EAC procurement can only be applied to project electricity use or district energy use. Captured bio-methane can only be applied to project fuel use.

Table 1. Points for Renewable Energy Procurement

Points	Tier 1		Tier 2		Tier 3	
	BD+C (Except CS)	CS	BD+C (Except CS)	CS	BD+C (Except CS)	CS
1	2%	1%	10%	3%	35%	10%
2	5%	3%	20%	6%	70%	20%
3	10%	5%	30%	9%	100%	30%
4	15%	10%	40%	12%		
5	20%	15%	50%	15%	-	-

GUIDANCE

Behind the Intent

Renewable energy criteria in v4.1 has been expanded to recognize the variety of procurement strategies that help to add renewable energy to the grid. Renewable energy generation can contribute to greenhouse gas emission reductions and offer local environmental benefits by reducing air pollution and increasing resilience. Renewable energy produced on-site protects projects from energy price volatility while reducing wasted energy lost in transmission.

Additionally, the voluntary market can be an effective catalyst for encouraging energy generators and utility companies to develop clean energy sources and help address climate change. Purchasing off-site renewable energy allows buildings that use nonrenewable power to create market demand for renewable energy and support the development of renewable infrastructure.

Beta Update

This new credit combines Green Power and Renewable Energy Production into one credit, recognizing the wide spectrum of renewable energy procurement. The credit adds new categories of renewables and updates performance requirements. The credit structure incentivizes self-supply of renewable energy and development of new renewables and provides further opportunities for building and portfolio owners to select the renewable procurement strategies that are most appropriate for the project application.

Step-by-Step Guidance

Step 1. Explore opportunities for renewables procurement.
(See Further Explanation, *Renewable Resource Considerations*)

- Step 2. Compare requirements for renewable energy systems and off-site methods of procurement.
Carefully evaluate the space requirements (for on-site systems), costs, financial incentives, and efficiencies for each potential renewable technology or contract.

- ▶ Local funding, financing, and incentives for renewable generation projects may be available for certain technologies and may be a significant factor. When considering funding options, ensure that the terms of the contract will address all renewable attributes to be retained by the project.
- ▶ For on-site systems, excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering). The building owner receives the market rate, however, and cannot charge a premium for the renewable energy. In effect, the grid serves as a storage system and frees the project from hosting a storage system on site. Alternatively, project teams may consider including a storage system to increase resiliency, facilitate further control of building energy costs, and reduce renewable energy curtailment.
- ▶ Tying into an existing community system or creating a community system may lower cost barriers through economies of scale, because unit costs may decrease as system sizes increase. Community systems can also take advantage of time-shifted demand: one building that is occupied during the day and another building that is occupied at night could both take advantage of the same biofuel-fired heating system. Ensure that the terms of the community renewable energy contract will address all renewable attributes to be retained by the project.
- ▶ For buildings that are part of a portfolio of buildings, renewable energy may be available from a third-party system, or the project team may enter an arrangement in which a third party owns a system that serves the project. In such cases, project teams must take additional steps to ensure that the arrangement continues for the contract period required in the credit, and that the renewable attributes are retained throughout the duration of the contract.
- ▶ Undertake a cost-benefit analysis to understand the financial and environmental benefit of all available options.
- ▶ When considering any financing mechanism or contract for renewable energy procurement, project teams must ensure that renewable attributes associated with the renewable energy generation will be retained by the project (See Further Explanation, *Energy Attribute Certificates*).
- ▶ Consider the additional criteria required to demonstrate EAp Minimum Energy Performance and/or EAc Optimize Energy Performance credit for Tier 1 or Tier 2 renewable energy using the ASHRAE 90.1 Appendix G Performance Rating Method or approved equivalent standard. Note that some Tier 1 and Tier 2 systems do not qualify for additional credit under EAp Minimum Energy Performance and/or EAc Optimize Energy Performance (See EAp Minimum Energy Performance, [Further Explanation, Applying Renewable Energy Savings](#)):

Step 3. Set target for renewable energy procurement.

Select one or more procurement strategies, for a total of up to 5 points. Each procurement strategy must meet or exceed the minimum target based on the percentage of total site energy use specified in the credit language.

To establish the target renewable energy system size for the project, estimate the annual site energy use for the project (See [Further Explanation, Total Site Energy Use](#)).

Review credit point thresholds and establish the renewable procurement goals for the project.

Step 4. Finalize Renewable Energy Procurement.

Purchase and install the renewable energy systems, and/or finalize the contract to procure renewable energy or Energy Attribute Certificates (EACs). Assure that the contract is signed by both parties, and that the contract terms confirm all credit requirements.

For on-site renewable energy system(s), review the contract to confirm that the renewable system(s) are scheduled to be operational at the time of building occupancy. On-site renewable systems must also be commissioned per the requirements of LEED EA prerequisite Fundamental Commissioning and Verification and EA credit Enhanced Commissioning, as applicable.

For Tier 2 renewable energy procurement, review the contract to confirm that the renewable system(s) are scheduled to be operational within two years of building occupancy.

Further Explanation

Calculations

Percentage of renewable energy purchased

Use Equation 1 to determine the percentage of renewable energy generated or purchased. Calculate the percentage of renewable energy generated or purchased for each Tier.

Equation 1: Percentage of energy generated or purchased

$$\% \text{ energy generated or purchased} = (\text{annual quantity of renewable energy in kWh}) / (\text{annual building site energy use in kWh})$$

Use Equation 2 to estimate annual energy use offset by captured biomethane:

Equation 2: Annual energy usage offset by captured bio-methane

$$\% \text{ fuel energy use offset} = (\text{annual quantity of captured bio-methane procured in kBtu}) / (\text{annual building site energy use in kBtu})$$

Use Equation 3 to calculate the number of points achieved for Tier 2 and Tier 3 when the contract term is different than ten years, and/or when procuring renewable energy from multiple Tiers.

Equation 3: Proration of Off-site renewables

Prorated Annual Renewable Percent = (Contract Term (in years)/ 10 years) x Annual Renewable Energy Generation / Annual Building Site Energy Consumption

Where:

- Annual Renewable Energy Generation and Annual Building Energy Consumption use consistent units of site energy (e.g. kWh/year)
- For a one-time purchase, contract term is listed as “one year”, and annual renewable energy generation is reported as the total renewable energy generation.

Total Site Energy Use

Total site energy includes all electricity, fossil fuel, district thermal energy, and other site energy delivered to the project.

Projects complying with EA prerequisite Minimum Energy Performance using energy simulation (i.e. Normative Appendix G Performance Rating Method, Section 11 Energy Cost Budget Method, or approved equivalent standard) must use the modeled total site energy documented for the Proposed design before excluding renewable energy as the basis of the renewable energy credit calculations (See Further Explanation, Example 1).

Projects complying with EA prerequisite Minimum Energy Performance through a prescriptive path must use the most recent building Energy Use Intensity by Property Type data from the ENERGY STAR Portfolio Manager Technical Reference to calculate total annual site energy for the renewable energy credit calculations. The site energy must be broken down into electric and non-electric consumption. This is calculated using the total source energy and total site energy reported in the ENERGY STAR Portfolio Manager Technical Reference, and the US source-to-site ratios for electricity and natural gas. If the project is all-electric with no combustion sources for heating, water heating, cooking, etc. in the project and base building, the total site EUI value may be considered as electricity. This energy source

calculation is included the LEED v4.1 Renewable Energy Calculator (See Further Explanation, Example 2).

Sum of points from multiple procurement sources

Using Table 1, add up the applicable points from equation 1, for on-site and off-site renewable energy; and equation 2 for captured bio-methane as applicable, for a total not to exceed 5 points.

This summation is performed in the LEED v4.1 Renewable Energy Calculator. For points documented in Tier 2 and Tier 3, total annual electricity procurement is limited to the total proposed annual site energy consumption from electricity and district thermal energy, excluding any Tier 1 electric generation.

Renewable Resource Considerations

The renewable energy credit seeks to increase overall demand for renewable energy and the use of grid-source renewable energy projects, with the goal of supporting broader grid-scale decarbonization. Criteria rewards renewable energy investments that have a high probability of causality (i.e. support development and installation of new renewables) and demonstrate long-term commitment. Project teams should follow a hierarchy for selecting renewable energy sources to meet credit requirements:

- ▶ First, on-site generation;
- ▶ second, local generation, such as community solar or wind, in instances where it will have a beneficial decarbonizing impact;
- ▶ third, offsite generation projects with high probability of causality, e.g. power purchase agreements;
- ▶ fourth, renewables from an existing renewable energy project, e.g. utility green tariff or direct access to wholesale markets
- ▶ last, energy attribute certificates (EACs)

The U.S. EPA's [Guide to Purchasing Green Power](#) provides additional information on the process of and strategies for procuring renewable energy.

Tier 1 On-Site Renewable Energy System Considerations

On-site renewable energy generation, when combined with careful consideration of building energy time of use and grid peak demand, and storage in some grid regions, can reduce annual greenhouse gas emissions, increase building resilience, and support effective grid management.

Eligible On-Site Renewable Energy Systems

To qualify as an on-site system, the renewable energy must be generated on-site from renewable sources produced at the building or contiguous campus site.

Allowable sources for on-site renewable energy include the following:

- ▶ Photovoltaic
- ▶ Solar thermal
- ▶ Wind
- ▶ Biofuel harvested on-site and used on-site for thermal and/or electric generation, excluding:
 - Combustion of municipal solid waste
 - Forest biomass waste, with the exception of the following harvested on site: clean urban wood waste, invasive species, habitat restoration programs, clean industrial wood waste (pallets), and tree tops left over from logging operations.
 - Wood coated with paints, plastics, or laminate

- Wood treated for preservation with materials containing halogens, chlorine compounds, halide compounds, chromated copper arsenate, or arsenic; if more than 1% of the wood fuel has been treated with these compounds, the energy system is ineligible
- ▶ Geothermal energy, such as electricity or heat generated from subterranean steam or hot water, excluding any geothermal energy used in conjunction with a vapor compression cycle.

Examples of on-site renewable energy generation include:

- ▶ A photovoltaic array located on the project site.
- ▶ A wind tower located on a contiguous campus owned by the same entity as the project building.
- ▶ Landfill gas processed in digesters on a contiguous campus owned by the same entity as the project building and used to produce thermal energy in the project building.

Note: earlier versions of LEED allowed some biofuels produced off-site to qualify as on-site renewable energy. However, based on the clarifications provided in ASHRAE 90.1-2016 for on-site renewable energy, and the clearer distinction between on-site and off-site renewable energy in LEED v4.1, biofuels are only considered on-site renewable systems when the renewable source is harvested on site, and used for on-site generation of electric or thermal energy.

Only usable energy generated from the on-site renewable system shall be considered towards the on-site renewable energy contribution. Usable energy is defined as the output energy from the system less any transmission and conversion losses, such as standby heat loss, losses when converting electricity from DC to AC, or waste heat in a cogeneration system that is exhausted to the atmosphere during periods of low thermal demand. Excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering) when Energy Attribute Certificates (EACs are retained by the project). Net metered electricity may count toward EA Renewable Energy credit up to 100% of annual electricity and district energy use.

A project team should use web resources and other tools available to determine the feasibility of renewable systems, given the project site's climate, context, and infrastructure. Consider the features of the site, such as solar availability, wind patterns, and other renewable energy sources, and any seasonal or daily variations in its supply. Certain project types may have special opportunities: office or university campuses typically have available land, for example, and warehouse projects may have large roof areas.

Match the project's energy needs with renewable energy output when selecting a renewable system. For example, a sunny site is a good candidate for solar thermal hot water, but this type of renewable resource is most cost-effective if the building has a constant demand for hot water. Accordingly, a hotel or a multifamily project may be a better match for a solar thermal hot water system than an office complex.

Daily and seasonal variations in loads also factor into the investigation of renewable energy. For example, a residential project with low daytime electricity demand may require battery storage to benefit from a photovoltaic (PV) array; an office building with high daytime demand may not.

On-site renewable systems must be installed and commissioned prior to the final LEED construction phase project submission to qualify for on-site renewable energy generation credit.

Tier 2 New Off-Site Renewables

Tier 2 off-site renewables are defined as those that have come online within the last five years or are contracted to be operational within two years of building occupancy.

Age of the generation asset(s) is assessed at the beginning of the contract, and the generation asset(s) retain these attributes for the duration of the contract or lease term. Therefore, a contract initiated for a new renewable asset for a portfolio of projects with EACs assigned to the project building would be considered "New renewable" even if the contract period initiated more than five years prior to project

occupancy. EACs assigned to the project shall be from the remaining contract duration on or after the year of project occupancy.

Community renewable energy cooperatives, larger-scale investments, such as direct, voluntary purchases in the form of power purchase agreements (PPAs), virtual PPAs, or renewable energy investment trusts, qualify as new off-site renewables provided documentation demonstrates that they meet the criteria described above. Contracts for these investments must indicate the specific system used to generate the renewable energy, with sufficient information available to confirm the renewable system generation capacity and allocation of the EACs (see *Further Explanation, Renewable Attributes*). If the LEED project contracts for a one-time purchase or short-term contract less than ten years for EACs exceeding 100% of the project's annual electricity and district thermal site energy use, the EACs must be Green-e Energy certified or equivalent.

Investment in new off-site renewables creates new renewable energy supply and has the potential to displace energy and emissions from fossil fuel-powered generators, particularly in regions where the grid mix is a higher percentage of fossil fuels.

To qualify as a new renewable system, the contract length shall be a minimum of ten years, or the annual renewable energy generation shall be prorated based on the contract term length. A commitment to renew does not qualify as a new renewable resource.

Tier 3 Off-Site Renewables

Tier 3 off-site renewables are defined as those contracted from an existing renewable energy provider or off-site renewable systems that were contracted for the building after the renewable system came online and came online more than five years before building occupancy.

Existing off-site renewables, which may include utility green tariff programs or direct access to wholesale markets, may be more widely available depending on project location or budget. Investment in existing renewable resources and utility programs remains an important strategy for sustaining market demand for renewables and ensuring financial viability of existing projects.

The contract length shall be a minimum of ten years, or the annual energy renewable energy generation shall be prorated based on the contract term length. Alternatively, Tier 3 procurement only, project teams may show compliance with the ten year minimum contract term by demonstrating the following:

- ▶ The project has an executed contract for a minimum of one year, OR where contracts are not available per regulatory requirements, document that the project has been enrolled in the Green-e or equivalent utility tariff for a minimum of one month.
- AND
- ▶ The building owner must provide a signed letter of commitment indicating that the project will remain continuously enrolled in the 100% renewable Green-e or equivalent utility tariff, or alternate Green-e or equivalent procurement source for a minimum of ten years (or the number of years documented for credit if less than 10 years).

Energy Attribute Certificates

An Energy Attribute Certificate (EAC) is a transferrable certificate, record or guarantee that provides information about the environmental attributes of one megawatt hour (MWh) of electricity. Renewable generators can sell EACs together with the electricity (bundled), or separately from the electricity (unbundled). EACs are also referred to as Renewable Energy Certificates (RECs) or Guarantees of Origin (GOs).

Retirement of environmental attributes of renewable electricity generation is substantiated through EACs. The owner of the building seeking LEED certification shall contract for renewable energy and demonstrate that the EACs are retained, owned, or retired on behalf of the party who has financial or operational control over the building's electricity use. Note that while the previous versions of LEED allowed EAC arbitrage for on-site systems, LEED v4.1 requires the owner to retain the environmental attributes of renewable energy that is generated on-site in order to for the generation to qualify for credit.

Demonstrating Green-e Equivalency

Projects not using Green-e certified products must demonstrate equivalency to the [Green-e Energy standard requirements](#) in the following areas:. To be considered equivalent to Green-e Energy, the EACs retired on behalf of the LEED project must be

- ▶ Certified under an eco-label or similar program developed by an independent organization with transparent accounting process and standards in place
- ▶ Certified under an eco-label developed by an independent organization with transparent accounting processes and standards
- ▶ Eligible renewable energy source (see Green-e Framework for Renewable Energy Certification, Section IIIA, "Renewable Resource Types", and additional regional requirements as applicable, i.e. Appendix D: Green -e Renewable Energy Standard for Canada and the United States, Section II (Eligible Sources of Supply)
- ▶ Renewable resources are from projects that have come online within the last fifteen years
- ▶ Verifiable chain of custody
- ▶ Mechanism to prevent double-counting

The executed contract must specify the purchasing goals consistent with the Green-e equivalency requirements and be valid for the duration indicated in the credit documentation.

Captured Bio-methane

Biogas from landfills, sewage treatment plants, and farm methane capture that is captured off-site and delivered to the LEED project for combustion on-site is considered "captured bio-methane" in LEED v4.1. Captured bio-methane may be used to offset fuel energy usage and qualify as Tier 3 off-site renewable energy provided that the biogas contract meets the minimum requirements for contract length. For procurement of captured bio-methane where a ten-year contract is not available, procurement shall be prorated based on the contract term length.

Retirement of environmental attributes of renewable gas is substantiated through certificates specific to gas production and use. The LEED project owner shall contract for renewable gas and demonstrate that the environmental attribute certificates are retained, owned, or retired on behalf of the LEED project.

Documenting Savings for EAp Minimum Energy Performance and EAc Optimize Energy Performance

Refer to EAp Minimum Energy Performance, [Further Explanation, Applying Renewable Energy Savings](#).

Rating System Variations

Core and Shell

CS projects shall determine total site energy use in the same manner as other BD+C projects, including the total site energy associated with the base building and future tenant fit outs (see Further Explanation, Total Site Energy Use).

Note:

The total site energy reference is aligned with EAc Renewable Energy Production from previous versions of LEED, which evaluates the Core and Shell renewable energy percentage based on total building energy use. The credit does not include provisions for calculating a lower “core and shell building energy” based on Core and shell floor area as was previously allowed in EAc Green Power and Carbon Offsets.

Project Type Variations

District Energy Systems (DES)

The method for documenting EA Renewable Energy credit must be consistent with the method for documenting EAp Minimum Energy Performance. If the project is using a prescriptive path to document EAp Minimum Energy Performance, OR applies ASHRAE 90.1 Appendix G to model district energy using identical purchased energy rates (i.e. \$/unit of delivered energy) and Greenhouse gas emissions factors (i.e. lb CO₂e emissions per unit of delivered district energy) in the Baseline and Proposed case for EAp Minimum Energy Performance, renewable electricity and EACs may be applied to offset the site energy use of the district energy (e.g. district steam, district hot water, district chilled water) in EAc Renewable Energy.

If the project documents a performance improvement for the district energy system within EAp Minimum Energy Performance using the LEED District Energy Guidance Path 2 or 3, the modeled results in the EAp Minimum Energy Performance will indicate the estimated natural gas, electricity, and/or other fuels associated with the district energy generation; and district energy will not be reported as an energy source in the modeled results. In this case, renewable electricity and EACs may only be applied to electricity (either in the building or associated with the district energy system) and captured bio-methane may only be applied to the fossil fuel or other fuel used in the building or associated with the district energy systems.

Required Documentation

Documentation
LEED v4.1 Renewable Energy Calculator
Renewable system rated capacity
Calculations to determine energy generated
Confirmation of renewable attribute ownership (for on-site systems owned by building owner) or contract indicating duration and renewable attribute ownership (for on-site systems owned by a third party)
Contract indicating percentage ownership, lease, or allocation of Tier 2 off-site renewable system
Purchase letter or contract of commitment showing renewable energy for targeted point threshold
Eco-label documentation showing Green-e Energy equivalency if not Green-e certified

Exemplary Performance

Tier 1: renewable energy generation meets or exceeds 25% of total site energy use.

Tier 2: renewable energy procurement meets or exceeds 60% of total site energy use.

Exemplary performance is not available for Tier 3 renewable energy procurement.

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Investments in renewable energy throughout the building life cycle can help reduce building greenhouse gas emissions and improve the building's energy performance score, increase market demand for renewables, and support the growth and financial feasibility of new renewable energy projects.

EA Credit: Enhanced Refrigerant Management

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To eliminate ozone depletion and global warming potential, and support early compliance with the Montreal Protocol, including the Kigali Amendment, while minimizing direct contributions to climate change.

REQUIREMENTS

NC, CS, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Option 1. No Refrigerants or Low-Impact Refrigerants (1 point)

Do not use refrigerants, or use only refrigerants (naturally occurring or synthetic) that have an ozone depletion potential (ODP) of zero and a global warming potential (GWP) of less than 50.

OR

Option 2. Calculation of Refrigerant Impact (1 point)

Comply with ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems, or USGBC-approved equivalent, as applicable to the project scope.

Develop and implement a refrigerant management plan that addresses leak detection, system retrofit, and end of life disposal for all HVAC&R systems containing more than 0.5 pound (225 grams) of refrigerant.

Select refrigerants that are used in heating, ventilating, air-conditioning, and refrigeration (HVAC&R) equipment to minimize or eliminate the emission of compounds that contribute to ozone depletion and climate change. The combination of all new and existing base building and tenant HVAC&R equipment that serve the project must comply with the following formula:

IP units	SI units
$\frac{\text{LCGW}}{P} + \frac{\text{LCOD}}{P} \times 10^5 \leq 100$	$\frac{\text{LCGW}}{P} + \frac{\text{LCOD}}{P} \times 10^5 \leq 13$
Calculation definitions for LCGWP + LCODP x 10⁵ ≤ 100 (IP units)	Calculation definitions for LCGWP + LCODP x 10⁵ ≤ 13 (SI units)
LCODP = [ODPr x (Lr x Life + Mr) x Rc]/Life	LCODP = [ODPr x (Lr x Life + Mr) x Rc]/Life
LCGWP = [GWPr x (Lr x Life + Mr) x Rc]/Life	LCGWP = [GWPr x (Lr x Life + Mr) x Rc]/Life
LCODP: Lifecycle Ozone Depletion Potential (lb CFC 11/Ton-Year)	LCODP: Lifecycle Ozone Depletion Potential (kg CFC 11/(kW/year))

LCGWP: Lifecycle Direct Global Warming Potential (lb CO ₂ /Ton-Year)	LCGWP: Lifecycle Direct Global Warming Potential (kg CO ₂ /kW-year)
GWPr: Global Warming Potential of Refrigerant (0 to 12,000 lb CO ₂ /lbr)	GWPr: Global Warming Potential of Refrigerant (0 to 12,000 kg CO ₂ /kg r)
ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 lb CFC 11/lbr)	ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 kg CFC 11/kg r)
Lr: Refrigerant Leakage Rate (2.0%)	Lr: Refrigerant Leakage Rate (2.0%)
Mr: End-of-life Refrigerant Loss (10%)	Mr: End-of-life Refrigerant Loss (10%)
Rc: Refrigerant Charge (0.5 to 5.0 lbs. of refrigerant per ton of gross AHRI rated cooling capacity)	Rc: Refrigerant Charge (0.065 to 0.65 kg of refrigerant per kW of AHRI rated or Eurovent Certified cooling capacity)
Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated)	Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated)

For multiple types of equipment, calculate a weighted average of all base building HVAC&R equipment, using the following formula:

IP units	SI units
$\frac{[\sum (LCGWP + LCODP \times 10^5) \times Q_{unit}]}{Q_{total}} \leq 100$	$\frac{[\sum (LCGWP + LCODP \times 10^5) \times Q_{unit}]}{Q_{total}} \leq 13$

Calculation definitions for [$\sum (LCGWP + LCODP \times 10^5) \times Q_{unit}$] / Qtotal ≤ 100 (IP units)	Calculation definitions for [$\sum (LCGWP + LCODP \times 10^5) \times Q_{unit}$] / Qtotal ≤ 13 (SI units)
Qunit = Gross AHRI rated cooling capacity of an individual HVAC or refrigeration unit (Tons)	Qunit = Eurovent Certified cooling capacity of an individual HVAC or refrigeration unit (kW)
Qtotal = Total gross AHRI rated cooling capacity of all HVAC or refrigeration	Qtotal = Total Eurovent Certified cooling capacity of all HVAC or refrigeration (kW)

RETAIL NC

Meet Option 1 or 2 for all HVAC systems.

Stores with commercial refrigeration systems must comply with the following.

- ▶ Use only non-ozone-depleting refrigerants.
- ▶ Select equipment with an average HFC refrigerant charge of no more than 1.75 pounds of refrigerant per 1,000 Btu/h (2.72 kg of refrigerant per kW) total evaporator cooling load.

- Demonstrate a predicted store-wide annual refrigerant emissions rate of no more than 15%. Conduct leak testing using the procedures in GreenChill's best practices guideline for leak tightness at installation.

Alternatively, stores with commercial refrigeration systems may provide proof of attainment of EPA GreenChill's silver-level store certification for newly constructed stores.

GUIDANCE

Refer to the LEED v4 reference guide, with the following edits and additions:

Behind the Intent

While the Montreal Protocol provided for the phase-out of CFC and HCFC refrigerants due to their ozone depletion potential, it led the industry shift toward the usage of hydrofluorocarbons (HFCs), which are potent greenhouse gases. In the Kigali Amendment to the Montreal Protocol, adopted in 2016, 197 countries committed to cut the production and consumption of HFCs by more than 80 percent over the next 30 years.

Alternatives to HFCs have been developed to provide lower-GWP refrigerant options, although the availability of low-GWP refrigerants varies between applications. The transition to lower-GWP refrigerant options requires an increased focus on safety, as many "next generation" refrigerants can pose flammability and/or toxicity risks if not properly managed. ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems provides essential guidance to manufacturers, design engineers and operators who need to stay current with new air conditioning and refrigerating requirements. The selection of refrigerants and their operating systems should be based on a holistic analysis of multiple criteria, including safety, environmental impacts, energy efficiency and cost.

In addition to protecting human health, effective refrigerant management during processes of design, manufacturing, operation, systems servicing, and end of life helps to reduce global emissions. The majority of refrigerant emissions happen at end of life, so the effective disposal of refrigerants currently in circulation (including CFCs, HCFCs and HFCs) is essential. A robust refrigerant management plan provides a framework for building owners and operators to properly manage these chemicals and minimize direct contributions to climate change.

Step-by-Step

Step 1. Document compliance with ASHRAE Standard 15-2019

Review ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems. The standard establishes procedures for the safe design, construction, installation and operation of refrigerant systems. Note that Standard 15 must be used with ASHRAE Standard 34-2019 Designation and Safety Classification of Refrigerants, which describes a shorthand way of naming refrigerants and assigns safety classifications based on toxicity and flammability data.

Ensure that all refrigeration equipment installed in the project comply with ASHRAE Standard 15-2019 requirements for restrictions on refrigerant use (section 7), installation restrictions (section 8), design and construction of equipment and systems (section 9), operation and testing (section 10) and general requirements (section 11), as applicable to the project scope.

Step 2. Develop and implement refrigerant management plan

Develop and implement a refrigerant management plan. The plan must include a complete inventory of all refrigerants used in the project and address refrigerant leak detection, system retrofit, and proper end of life disposal for all HVAC&R systems containing more than 0.5 pound (225 gram) of refrigerant.

The plan must identify the individuals responsible for the care and maintenance of refrigerant systems, regular leak detection and monitoring, record-keeping and training of personnel necessary to execute the plan.

All contractors or service technicians performing work on any HVAC&R systems shall have access to the refrigerant management plan and, upon completion of the work, provide documentation that confirms continued compliance with the plan.

Step 3. Calculate refrigerant impact of proposed systems

Refer to the LEED v4 reference guide.

Step 4. Incorporate design criteria into project plans and specifications

Refer to the LEED v4 reference guide.

Further Explanation

Referenced Standards

- ▶ ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems

Required Documentation

Refer to the LEED v4 Reference Guide content, plus the following additions for Option 2:

- ▶ Confirmation that project HVAC&R systems comply with ASHRAE Standard 15
- ▶ Refrigerant management plan

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Enhanced Refrigerant Management: Effective refrigerant selection and management, especially at the point of disposal, is a critical strategy for addressing climate change and minimizing the release of building-related emissions into the atmosphere. Additionally, careful consideration of refrigerants used in HVAC&R equipment can improve performance and reduce operating costs throughout the building life cycle.